Project Title: Machine Learning-Driven Handover Optimization in LTE networks: Quality Signal Indicator, Classifier-Based Prediction Analysis and Hybrid Q-Learning.

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Abstract

The increasing complexity of mobile networks and the demand for seamless connectivity have made efficient handover management a critical challenge, particularly in LTE environments. This study presents a machine learning-driven approach to handover optimization, leveraging real-world drive test data to enhance network performance. The research integrates three key methodologies: Quality Signal Indicator (QSI) classification, supervised learning-based handover prediction, and hybrid Q-learning for optimization. First, QSI classification employs percentile-based thresholding and unsupervised clustering to assess wireless signal quality, categorizing it into four levels: Excellent, Good, Moderate, and Poor. Next, a comparative evaluation of ten supervised learning models, including Decision Trees, Random Forest, and XGBoost, is conducted for handover prediction. Performance metrics such as precision, recall, and F1-score highlight the effectiveness of these classifiers in identifying potential handover events. To optimize handover efficiency, a reinforcement learning-based Q-learning approach is implemented, refining handover decision-making by balancing exploration and exploitation strategies. The model integrates a reward function that penalizes unnecessary handovers while rewarding optimal transitions, ultimately reducing redundant handover instances from 97 to 71 within the dataset. This hybrid framework demonstrates a significant reduction in unnecessary handovers while maintaining seamless connectivity. The results indicate that machine learning and reinforcement learning can substantially enhance handover management, reducing network overhead and improving signal stability. The study lays the foundation for further research in adaptive, self-optimizing networks, with potential extensions to 5G and beyond. Future work will focus on incorporating additional parameters, such as latency and throughput, for a more holistic optimization strategy.

1. Introduction:

The increasing demand for seamless mobile connectivity, particularly in high-mobility environments such as highways and railways reflect the importance of optimizing handover mechanisms in wireless communication networks.

Handover, which is the process of transferring a user's connection from one base station to another. It is crucial for maintaining continuous service and real world relevance in ensuring quality user experience.

Traditional handover strategies based on fixed signal strength thresholds often result in unnecessary transitions or delayed decisions, negatively affecting network performance.

Therefore, this project is needed to explore modern, intelligent alternatives that can enhance handover efficiency and reliability.

The core problem lies in the inefficiency of traditional handover approaches, which do not adapt well to dynamic network conditions and can lead to increased latency and frequent disconnections.

There is an existing gap in leveraging real-time data and predictive models for proactive handover decisions. Most existing methods overlook the potential of advanced data analytics and machine learning.

Additionally, a major challenge encountered was the lack of publicly available datasets relevant to this study. To overcome this, a drive test was conducted to collect real-world data, which then required extensive pre-processing and cleaning before it could be used for model development.

The key goals of this project are: To utilize machine learning (ML) techniques for improving the handover decision-making process.To analyze wireless signal quality metrics such as RSRP, RSRQ, and neighboring cell conditions.To design a data-driven framework capable of predicting and optimizing handovers.

This project focuses on the analysis and development of a machine learning-based handover optimization model using real-world drive test data.

The scope includes wireless signal parameter analysis, data pre-processing, feature selection, and ML model training and evaluation.

Limitations include the scarcity of publicly available datasets, which necessitated the time-intensive process of data collection via drive testing. Moreover, computational constraints and the complexity of real-world signal variability posed additional challenges during data processing and model development.

2. LITERATURE REVIEW / RELATED WORK

2.1 EXISTING STUDIES

Handover management and Quality of Service (QoS) optimization in mobile networks, particularly in 4G and 5G, have been extensively studied. Traditional handover techniques rely on predefined thresholds for key performance indicators (KPIs) such as Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), and Signal-to-Interference-plus-Noise Ratio (SINR) [1]. While these methods are effective under static conditions, they often fail in dynamic environments, leading to issues such as unnecessary handovers (ping-pong effects) or delayed handovers, which degrade QoS [2].

To address these limitations, machine learning (ML)-based handover optimization techniques have gained traction. Supervised learning models such as Decision Trees, Support Vector Machines (SVM), and Neural Networks have been used to predict optimal handover decisions based on historical network data [3]. Deep learning techniques, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have been applied to spatiotemporal handover prediction, improving prediction accuracy but requiring extensive computational resources and large datasets [4].

Reinforcement learning (RL) has also been explored as an adaptive approach to handover optimization. Q-learning, a model-free RL algorithm, enables intelligent decision-making by learning optimal handover policies through interaction with the network environment [5]. Studies have shown that Q-learning can dynamically adjust handover thresholds, select optimal target cells, and minimize unnecessary handovers [6]. Deep Q-networks (DQN), an extension of Q-learning, further improve decision-making in high-dimensional state spaces [7].

However, existing ML and Q-learning-based approaches face several challenges. Many models require extensive training data and high computational resources, making real-time deployment difficult [8]. Furthermore, most studies rely on simulations rather than real-world network data, which limits their applicability in live network scenarios [9]. Another limitation is the lack of efficient state-space representation, which can lead to suboptimal policies and increased computational complexity [10].

2.2 COMPARISON WITH EXISTING WORK

This research builds upon existing ML and Q-learning-based handover optimization strategies by refining state representation, reward mechanisms, and feature selection techniques. Unlike conventional threshold-based handover mechanisms, this study leverages predictive modeling and reinforcement learning to enhance handover decision-making before network degradation occurs, improving user experience and reducing call drops [11].

One key innovation in this research is the integration of feature selection techniques tailored to handover decision-making. Many prior studies overlook the impact of feature selection on learning efficiency and accuracy. By incorporating domain-specific knowledge of 4G/5G mobility patterns, this study refines feature engineering, reducing computational overhead while improving model performance [12].

Additionally, unlike previous research that primarily relies on simulations, this study integrates real-world network data to validate the proposed ML and Q-learning-based handover models. By leveraging real measurement data, this research ensures that optimized handover policies are practical and applicable in live network environments [13].

The need for a new approach arises from the increasing complexity of mobile networks, where static handover mechanisms and naive ML/Q-learning models struggle to adapt to varying user mobility patterns and network load conditions. By integrating adaptive ML and reinforcement learning techniques with real-world validation, this research contributes to more efficient, intelligent, and scalable handover optimization in next-generation wireless networks [14].

3. System Architecture

To conduct this project, we require a dataset containing real-time signal measurements from both the serving cell and its neighboring cells. Since our focus is on LTE networks, acquiring such a dataset—particularly one that includes signal values and handover information—is challenging due to its rarity. To address this issue, we generated a dataset through drive testing. This section provides an analysis of the drive test methodology, dataset preparation, and overall project architecture.

3.1 Drive test

For the drive test, we used XCAL-M software connected to a Samsung Exynos chipset while traveling through a busy road in Dhaka. The location was chosen on the basis of condensed LTE base stations indicated by the OpenCellID website. The Xcal software recorded real-time signal values, providing essential network performance data. Additionally, we used an Arduino Uno with a Neo-7M GPS module to capture the user's speed and user's position ensuring accurate mobility analysis.

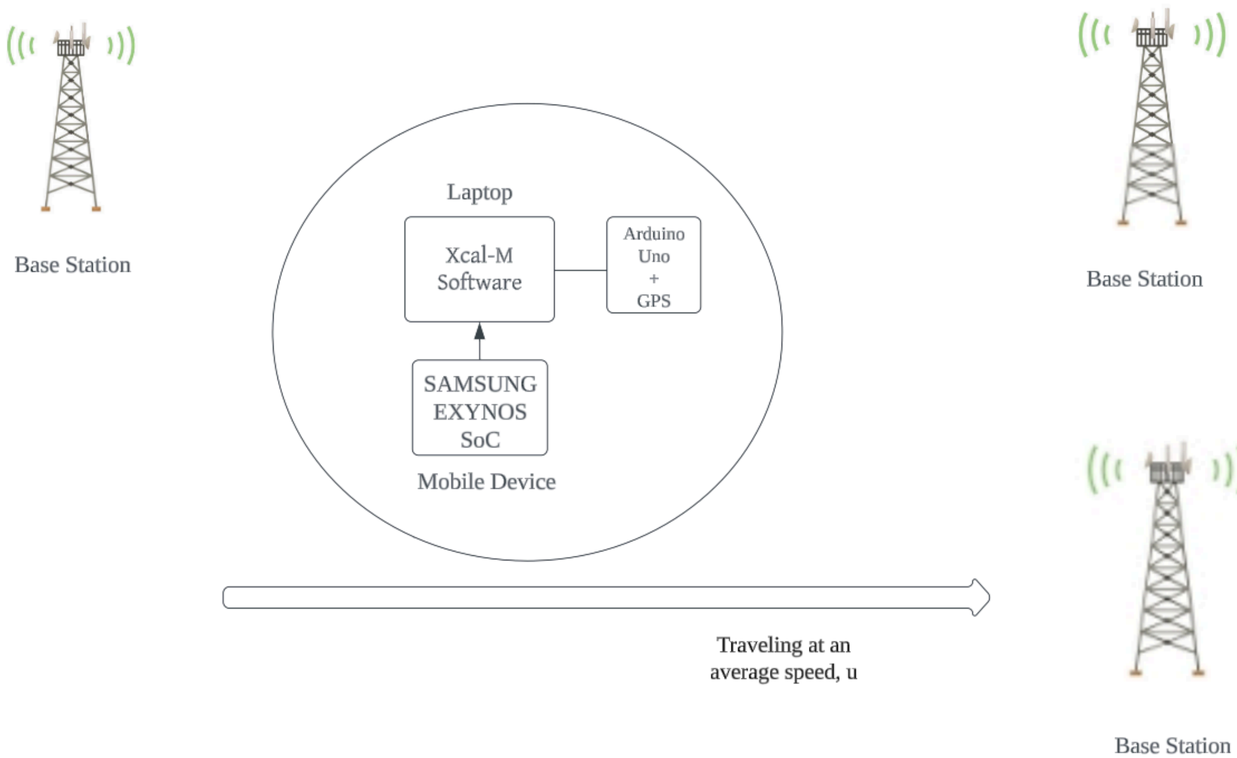
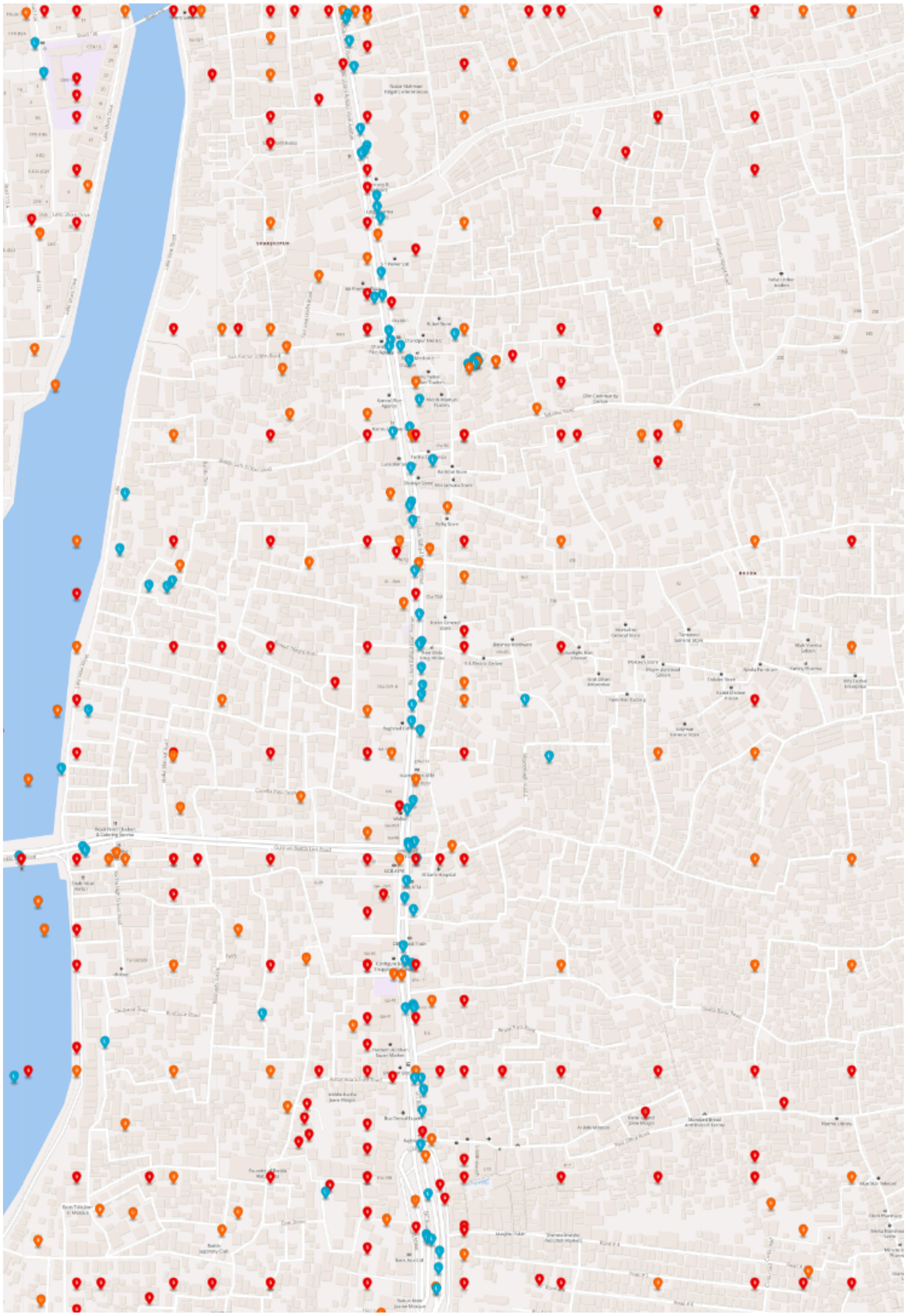


Fig 1: Drive Test



Fid 2: Open Cell ID website view of drive test location

The blue marked points indicate the LTE base stations. We travelled through this path to cover the LTE network as much as we can to get better handover statistics.

Fig 3: Drive Test Real Map

3.2 Data Preprocessing

We extracted signal values from LTE measurement reports in XCAL, while handover information was obtained from event statistics. Cell ID details were retrieved from the LTE summary table. Finally, we processed and structured the data into a CSV file for further analysis. We handled the missing values through linear interpolation and time to trigger cases were manually refined taking 320 ms duration.

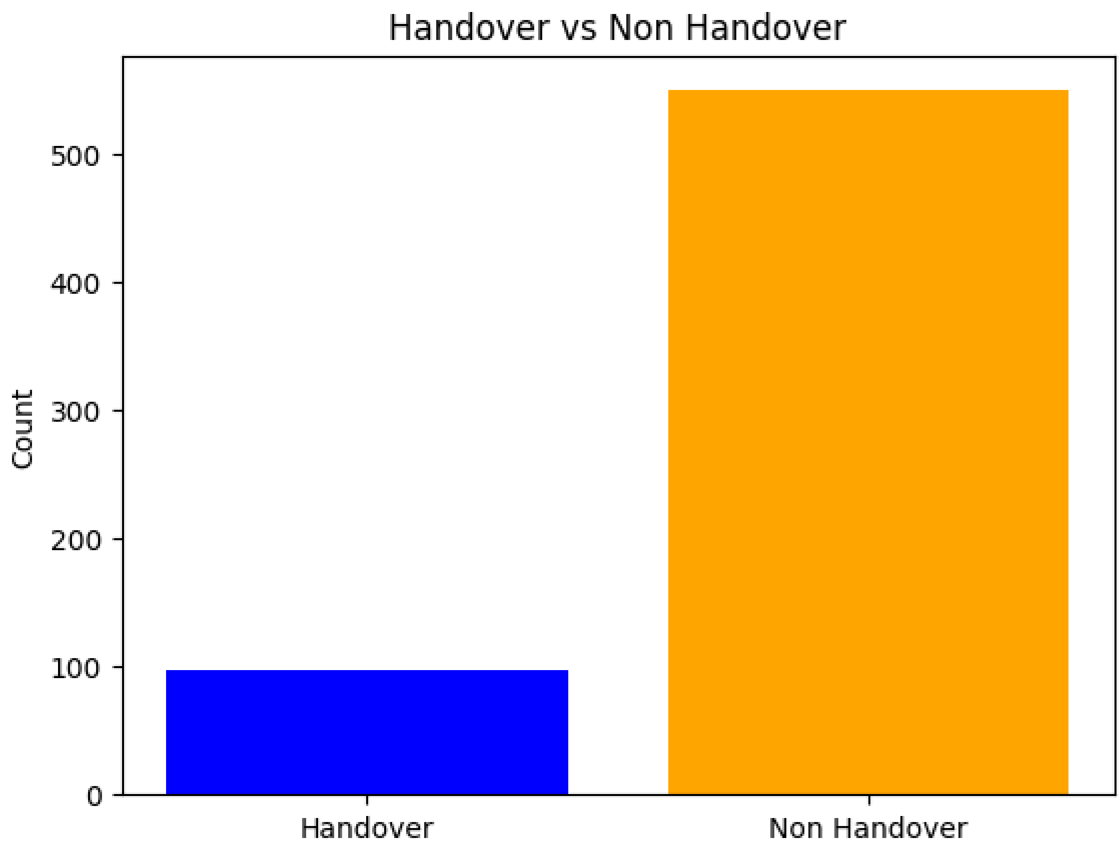


Fig 4: Label Counts

The dataset contains 10 features and 1 target variable. All of them are numerically valued columns. There are 647 data points and 94 handover instances, indicating imbalance in the dataset. Table-1 shows the feature titles and the target variable.

Features	Serving Cell Id
	Serving Cell RSRP
	Serving Cell RSRQ
	Serving Cell CINR
	Neighbor Cell Id
	Neighbor Cell RSRP
	Neighbor Cell RSRQ
	User Speed
	Latitude
	Longitude
Target Variable	Handover

Table 1: Dataset Columns

The correlation heatmap and the distribution of the numerical features are shown below in the figures.

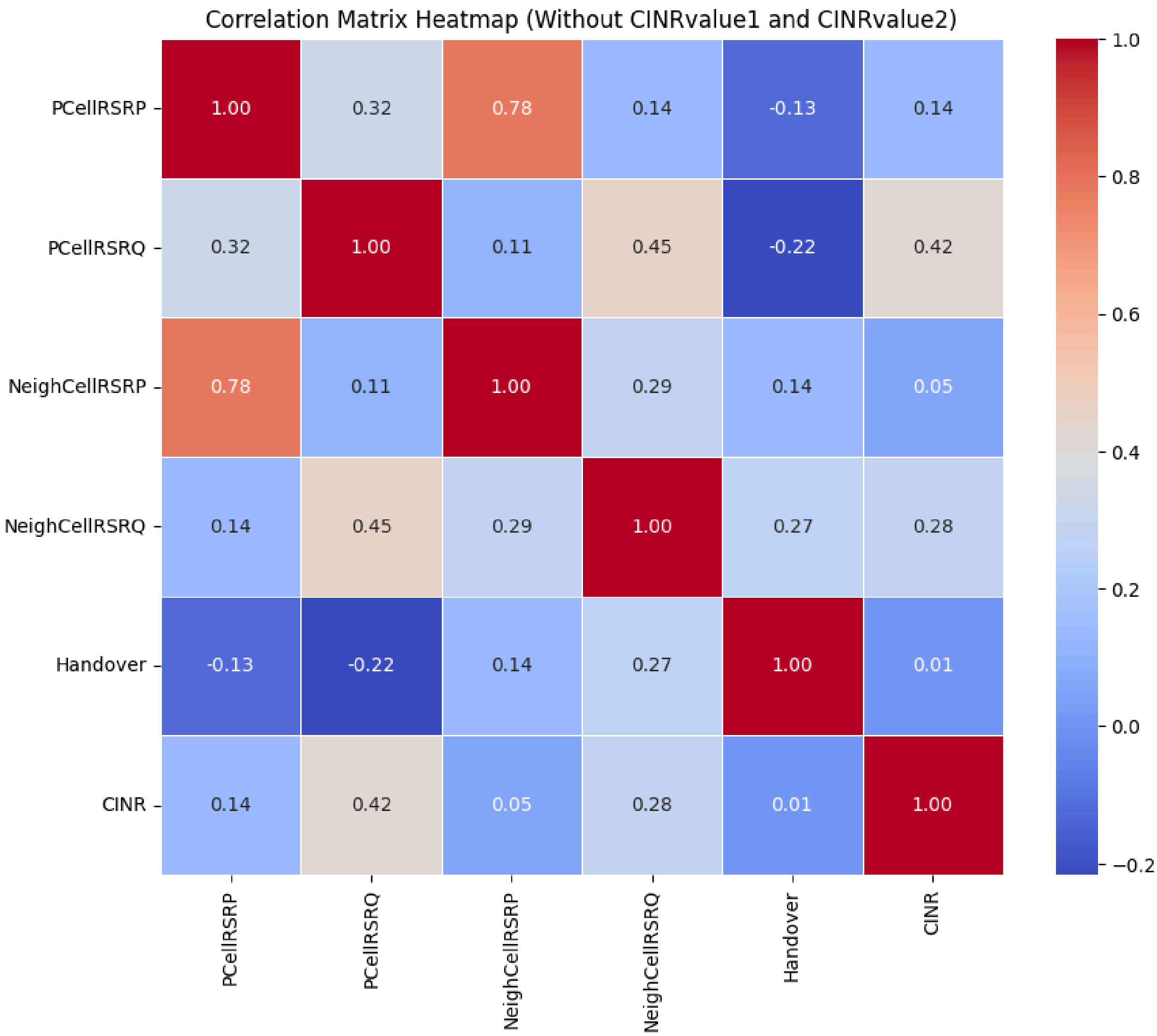
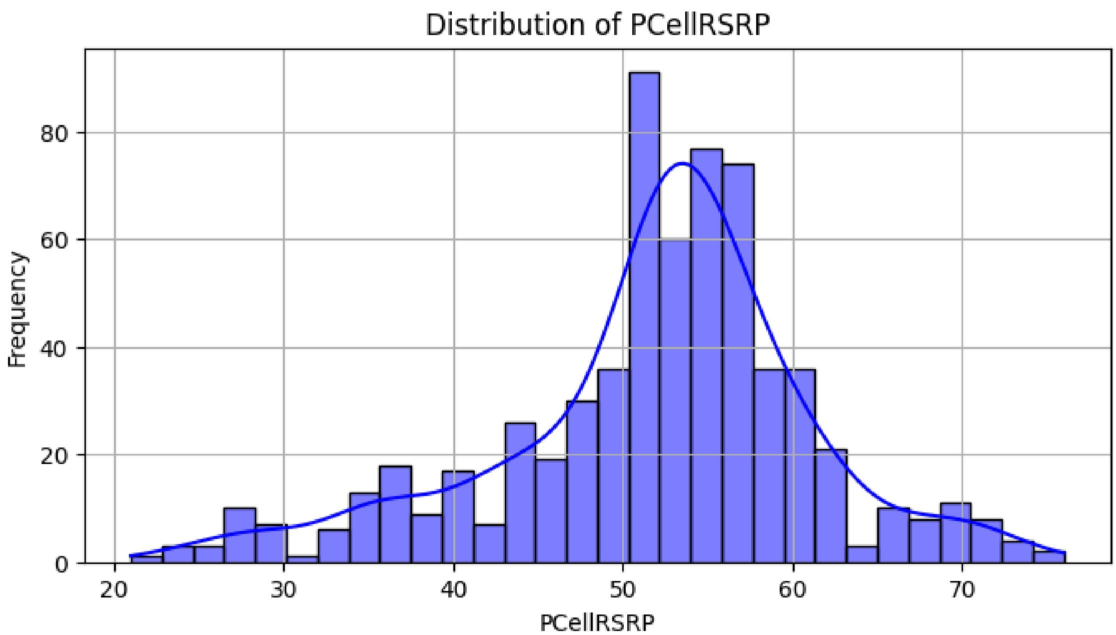
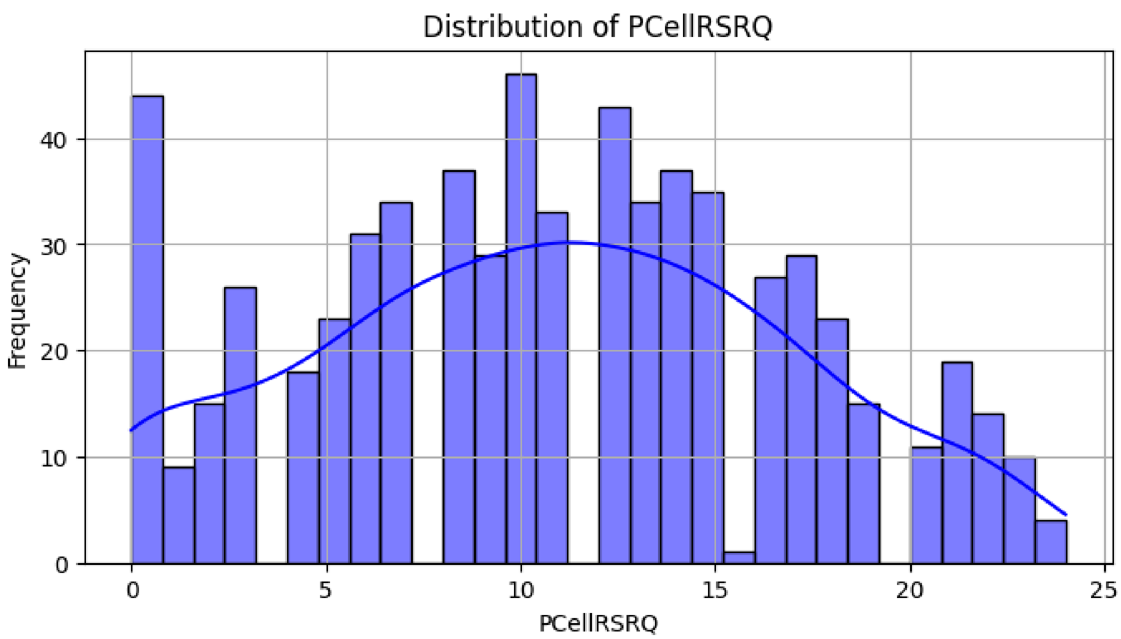
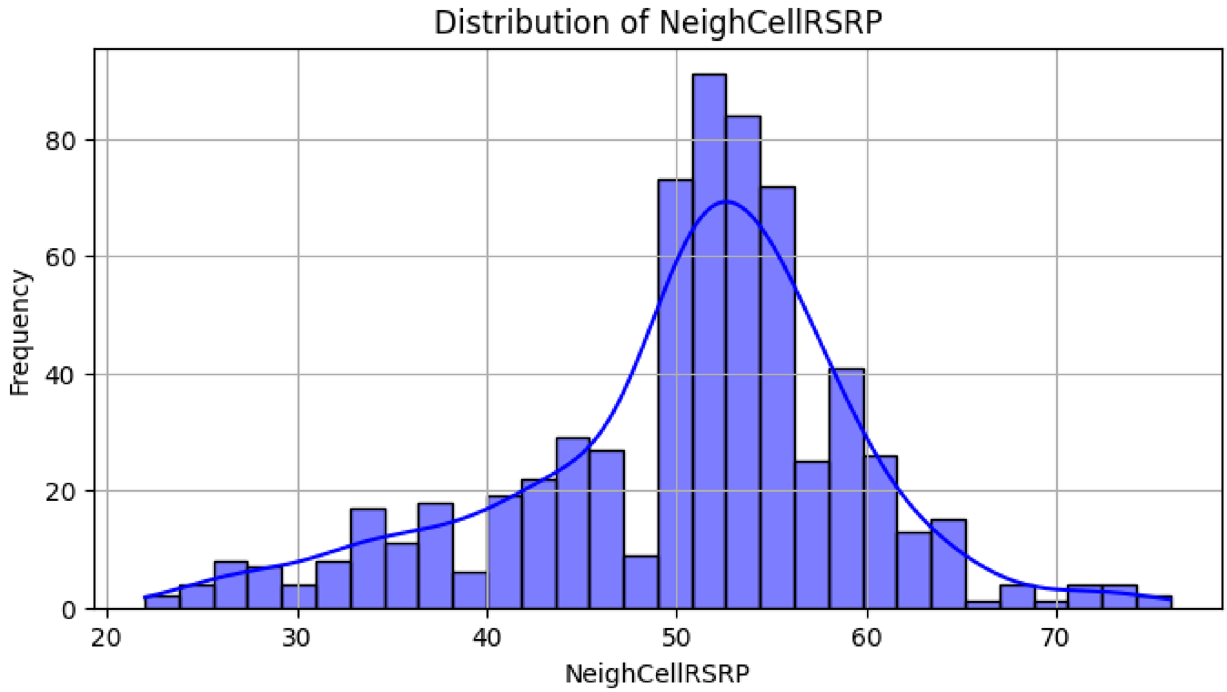


Fig 5: Correlation Matrix Heatmap of features.


Distribution of PCellRSRP
Frequency
PCellRSRP
0
20
40
60
80
20
30
40
50
60
70


Distribution of PCellRSRQ
Frequency
PCellRSRQ
0
10
20
30
40
0
5
10
15
20
25


Distribution of NeighCellRSRP
Frequency
0
20
40
60
80
20
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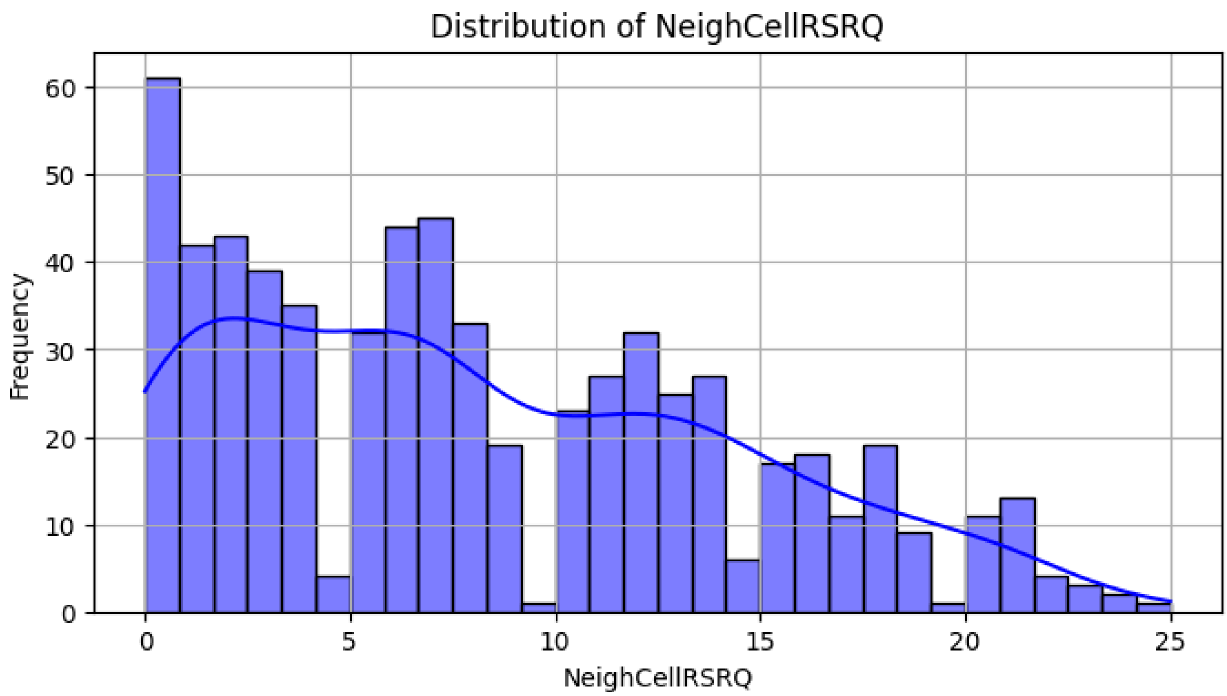
40
50
60
70
NeighCellRSRP


Distribution of NeighCellRSRQ
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NeighCellRSRQ

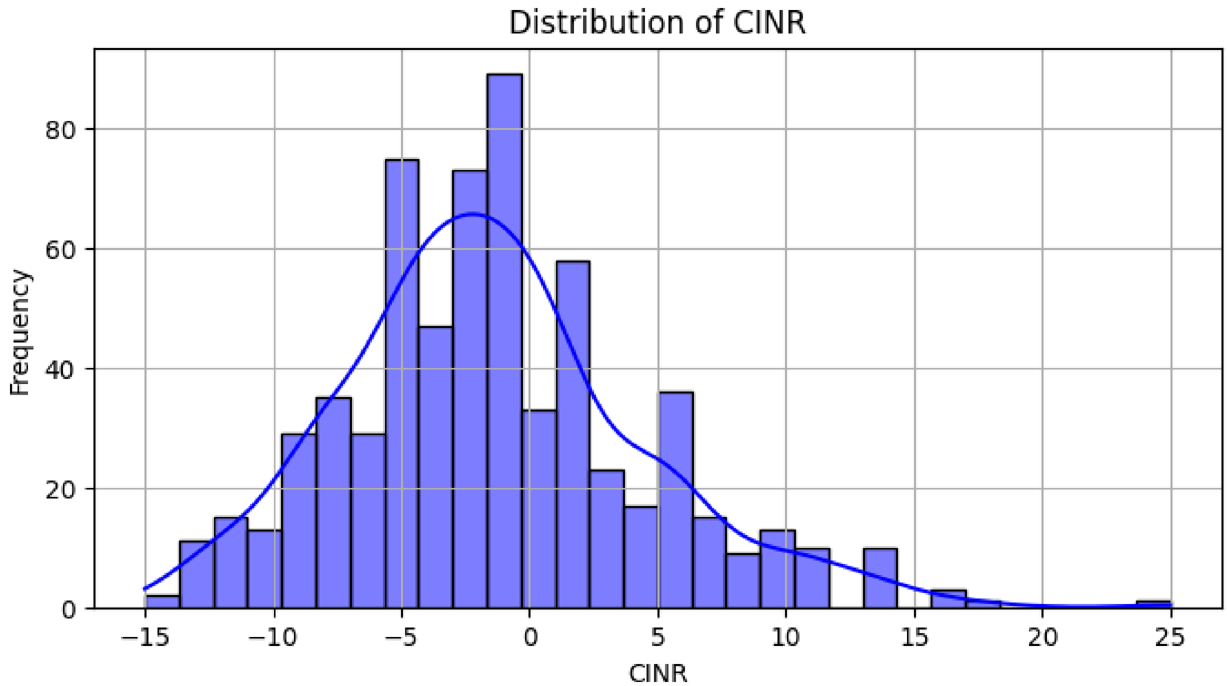

Fig 6: Distribution plots of features.

It is shown that the distribution follows Gaussian normal distribution.

Figure 7 and Figure 8 shows the relation between RSRP and RSRQ values for both serving cell and neighbor cell in scatter plot.

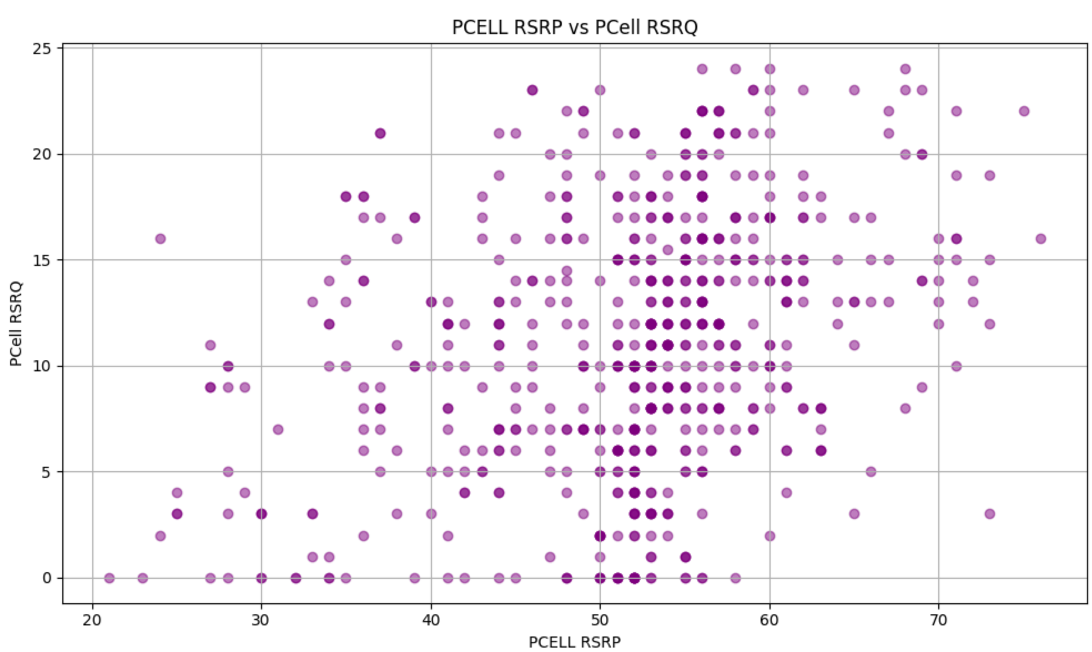

Figure 7: Scatter plot of Serving Cell RSRP and RSRQ

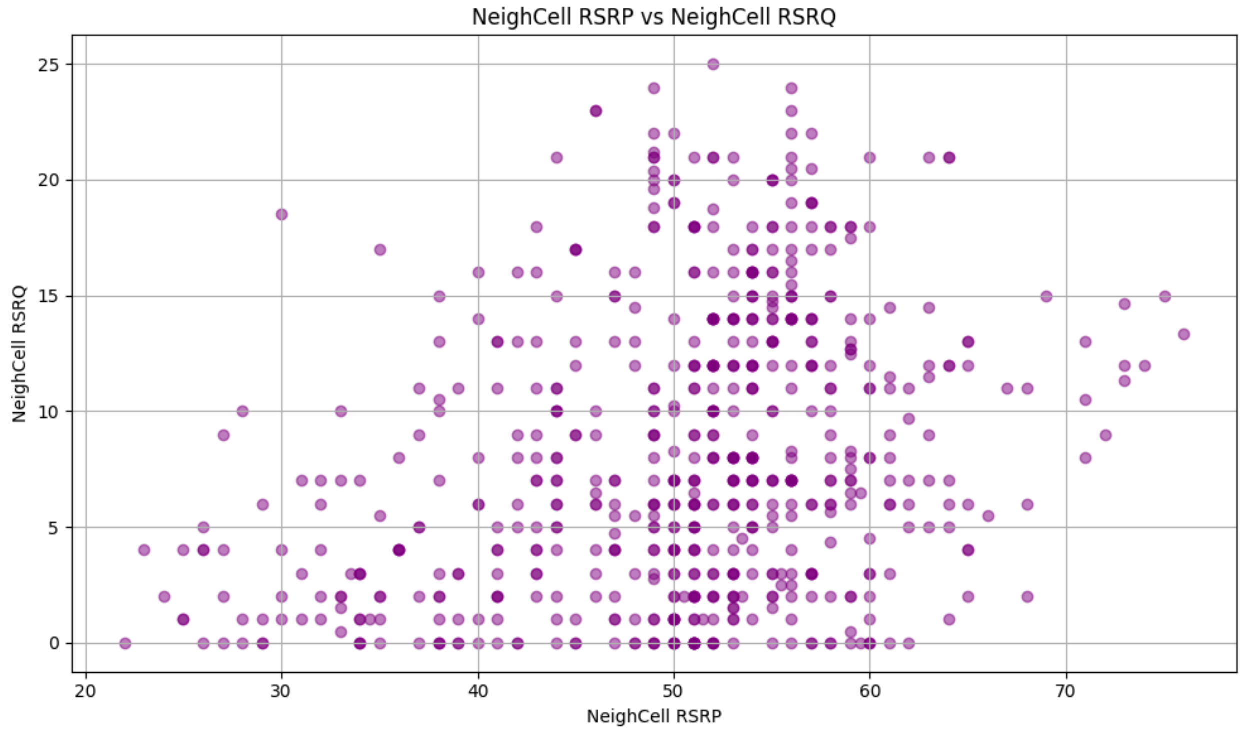

Fig 8: Scatter plot of Neighbor Cell RSRP and RSRQ

Figure 9 shows the number of handovers for serving cell IDs in the dataset.

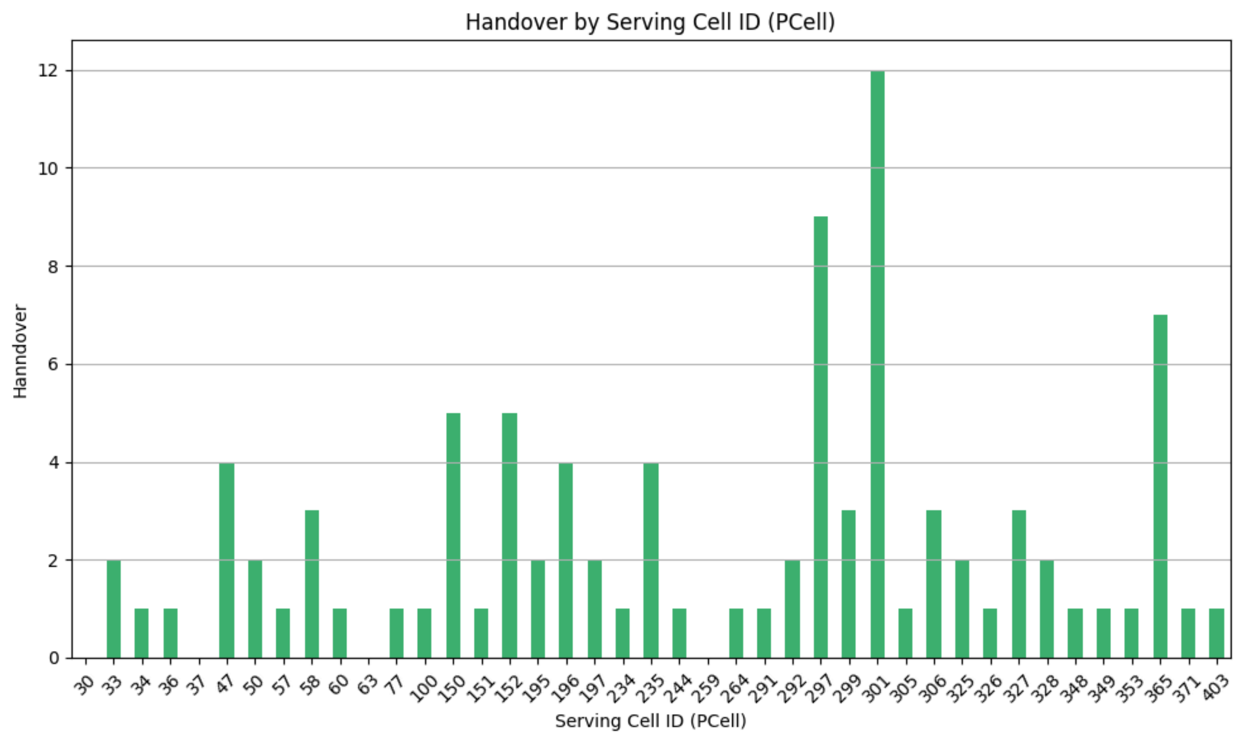

Fig 9: Handover frequency in serving cell IDs.

In the following sub sections, we discuss the project workflow based on the dataset we generated. Our study focuses on three key aspects: Quality Signal Indicator (QSI), Handover Prediction, and Handover Optimization, each of which will be elaborated on in the methodology section.

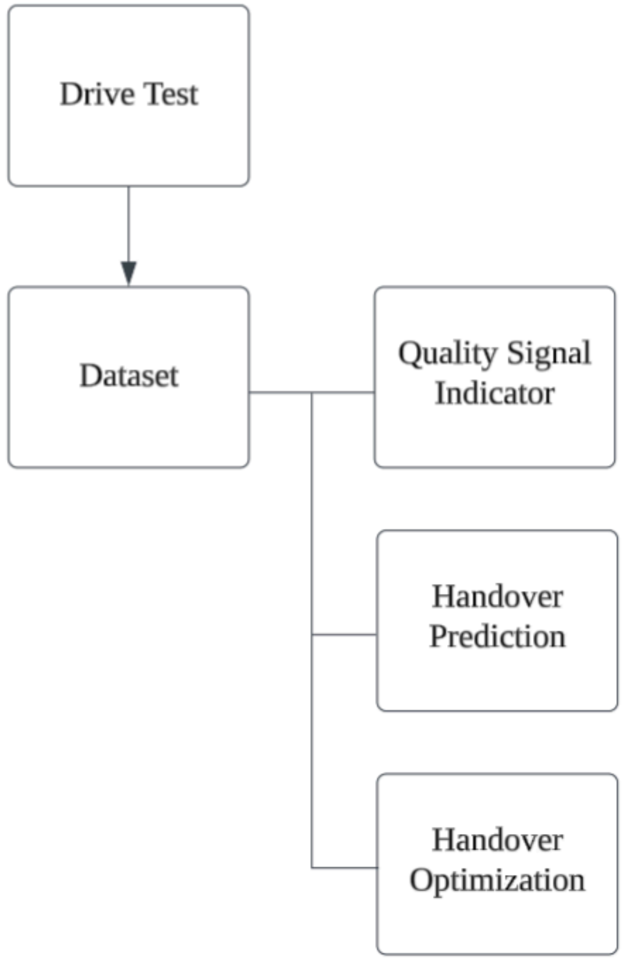

Fig 10: Project Overview

3.3 Quality Signal Indicator (QSI)

To assess network performance, we clustered signal quality into four categories: Excellent, Good, Moderate, and Poor, based on serving cell RSRP, RSRQ, and CINR values. The classification was performed using percentile-based thresholding, ensuring adaptive categorization across varying network conditions. We implemented this clustering through statistical analysis and threshold-based segmentation, forming a basis for signal quality assessment.

3.4 Handover Prediction

For handover prediction, we designed an ML-based architecture and implemented 10 supervised classification models to predict potential handover events. The classifiers were trained on labeled drive test data, incorporating signal trends and mobility parameters. The models were evaluated based on accuracy, precision, and recall to determine the most effective predictor of handovers.

3.5 Handover Optimization

To minimize unnecessary handovers, we employed Q-learning for reinforcement-based optimization. A Q-table was designed and iteratively updated to find optimal handover decisions, balancing network stability and mobility performance. The Q-learning model was trained and evaluated to ensure improved handover efficiency while maintaining service quality.

Each of these methodologies will be discussed in greater detail in the following section.

4. Methodology

This section provides a comprehensive analysis of the methodologies and algorithms employed in the three key subprojects: Quality Signal Indicator (QSI), Handover Prediction, and Handover Optimization. It details the implementation strategies and model architectures used in each approach. The section concludes with a discussion of the potential limitations of the proposed methodology.

4.1 Quality Signal Indicator (QSI) Analysis

4.1.1 Signal Quality Labeling and Categorization

The dataset provides real-time signal quality metrics, including Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), and Carrier-to-Interference-plus-Noise Ratio (CINR) for the serving cell. Initially, we applied manual threshold-based labeling, classifying signal quality into four categories: Excellent, Good, Moderate, and Poor. This labeling approach follows a self-supervised learning methodology, where predefined conditional thresholds segment the dataset, enabling structured categorization.

4.1.2 Model Training on Self Supervised Data

In this study, nine machine learning classifiers—Logistic Regression, Random Forest, K-Nearest Neighbors, Gradient Boosting, XGBoost, AdaBoost, Decision Tree, Naïve Bayes, and Support Vector Machine—were applied to self-supervised data for a multiclass classification task. The target variable, Quality Signal Indicator (QSI), was categorized into four distinct labels: Excellent, Good, Moderate, and Poor. These classifiers were trained on features such as RSRP, RSRQ, and SNR, aiming to develop a robust predictive model for signal quality assessment in telecommunications networks.

4.1.3 Dimensionality Reduction Using Principal Component Analysis (PCA)

To enhance computational efficiency and mitigate redundancy, Principal Component Analysis (PCA) was employed for dimensionality reduction. PCA transforms the original feature space into a set of orthogonal components, where the first few principal components retain the maximum variance of the data. This method reduces feature interdependence while preserving essential information, optimizing the dataset for further clustering analysis.

4.1.4 Clustering and Centroid Extraction

Following PCA, we visualized the signal categories using a scatter plot to analyze the distribution of signal quality clusters. From this plot, we extracted the centroids of each category, which served as the initial cluster centroids for K-means clustering.

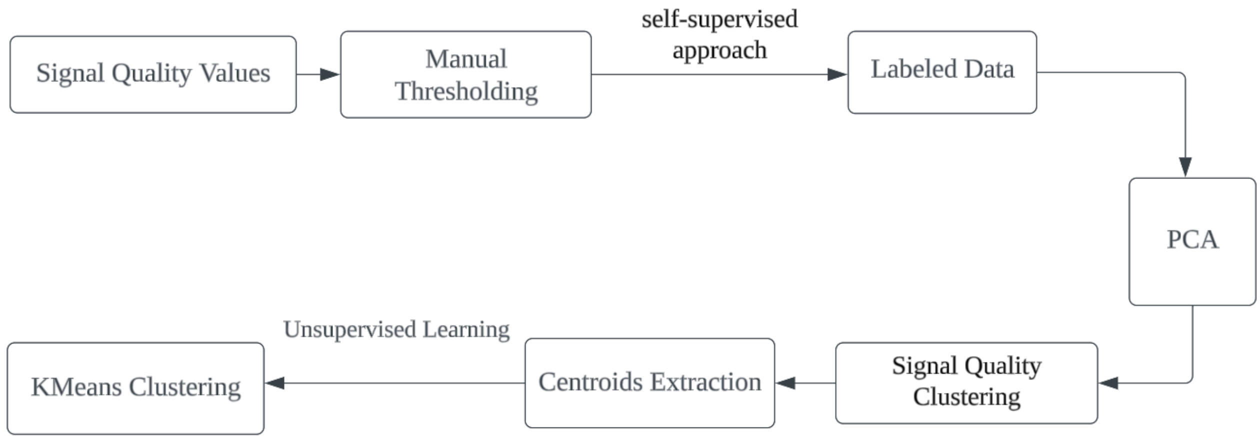

Fig 11: QSI workflow

4.1.5 Unsupervised Clustering with K-Means

K-means clustering, an unsupervised learning algorithm, was implemented to refine the signal quality classification. The algorithm operates iteratively by:

Initialization: Assigning predefined centroids extracted from the scatter plot.

Cluster Assignment: Each data point is assigned to the nearest centroid based on Euclidean distance.

Centroid Update: New centroids are computed as the mean of all points within a cluster.

Convergence: The process repeats until centroid positions stabilize, ensuring optimal cluster formation.

This clustering approach enhances signal quality classification by leveraging data-driven centroid refinement rather than relying solely on static threshold-based segmentation. The results of K-means clustering provide a robust and scalable model for network performance assessment.

4.2 Handover Prediction Model

4.2.1 Imbalance Handling and Dataset Preprocessing

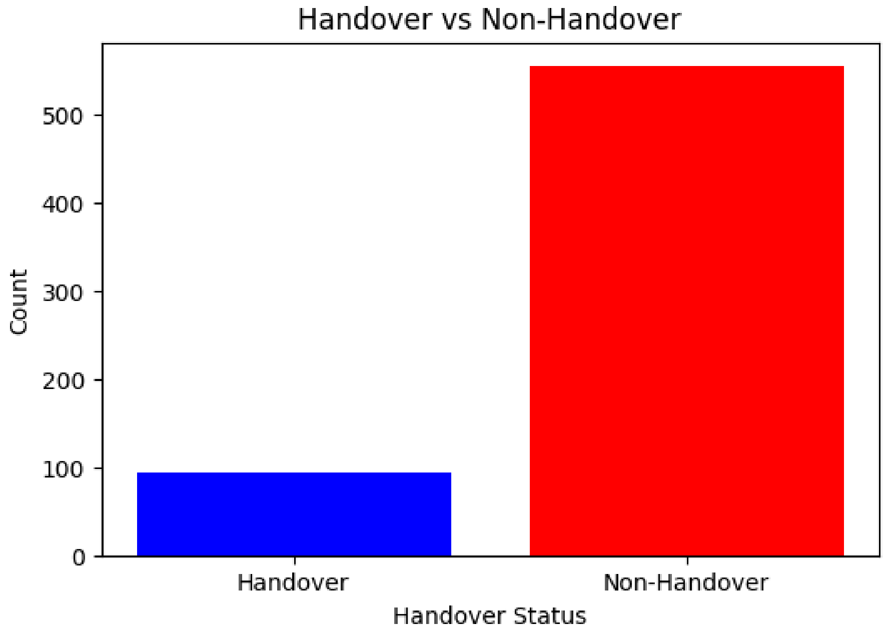

Fig 12: Imbalance Dataset

Due to significant class imbalance in the dataset, we designed a robust handover prediction architecture to enhance model performance. The dataset was initially split into training and testing sets, with further partitioning of the training set into majority (non-handover) and minority (handover) subsets. To address imbalance, we applied Random Undersampling (RUS) for the majority class and Synthetic Minority Over-sampling Technique (SMOTE) for the minority class.

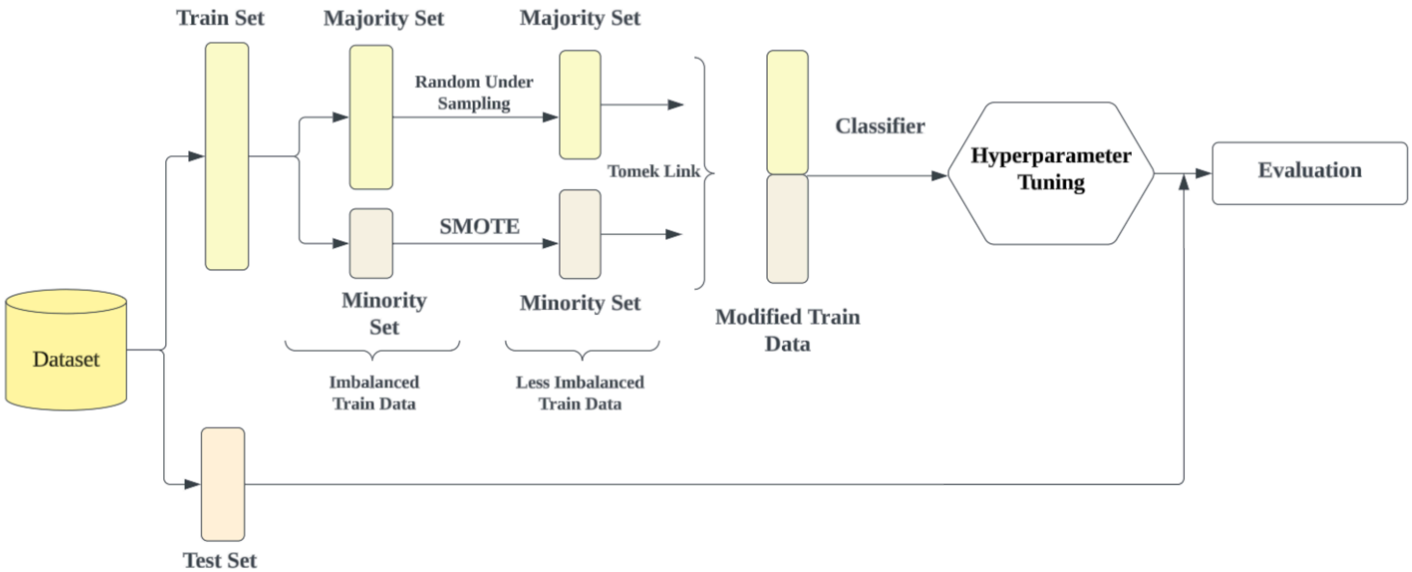

Fig 13: Proposed Architecture

Random Undersampling (RUS): This technique randomly removes samples from the majority class, reducing over-representation and preventing classifier bias.

SMOTE (Synthetic Minority Over-sampling Technique): SMOTE synthetically generates new minority class instances by interpolating between existing samples. It selects a minority sample, finds its nearest neighbors, and generates synthetic points along the feature space to improve class balance.

After rebalancing, we applied Tomek Links to further refine the dataset.

Tomek Link-Based Noise Reduction: Tomek Links are data pairs where one sample belongs to the majority class and the other to the minority class, with minimal inter-class distance. Removing the majority class instances in these pairs enhances class separation, improving classifier performance by reducing borderline noise.

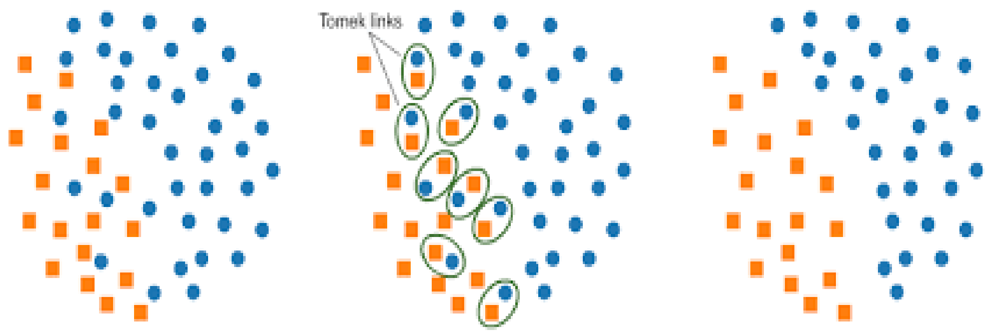

Fig 14: Tomek Link functionality (source: Kaggle)

4.2.2 Model Training

We trained 10 supervised classification models: Logistic Regression, Decision Tree, Random Forest, k-Nearest Neighbors (k-NN), Naïve Bayes, Support Vector Machine (SVM), AdaBoost, Gradient Boosting, CatBoost, and XGBoost. Each model underwent hyperparameter tuning using techniques such as Grid Search and Random Search, optimizing performance metrics like accuracy, precision, recall, and F1-score.

Finally, the models were evaluated on the test set, comparing their predictive capabilities against a baseline model to determine the most effective classifier for handover prediction.

4.3 Handover Optimization Using Hybrid Q-Learning

4.3.1 Hybrid Reinforcement Learning Framework

To optimize handover decisions, we implemented a hybrid Q-learning model that integrates Logistic Regression-based action selection with traditional Q-learning. This approach balances model-driven decision-making with reinforcement learning to minimize unnecessary handovers while maintaining signal quality.

4.3.1.1 State, Action, and Reward Definition

In the Q-learning framework:

State (S_t): Each state represents a data point containing real-time RSRP, RSRQ, and CINR values.

Action (a_t): The agent (User Equipment) can choose between handover or non-handover.

Reward Function (R_t): The agent receives a positive reward for an optimal handover and a negative reward (penalty) for an unnecessary or failed handover. The reward function is designed to minimize frequent handovers while maintaining network stability.

Condition	Action	Reward/Penalty	Explanation
Action = 1 (Handover)	Handover	-	Decision to perform a handover
NeighCell RSRP ≤ PCell RSRP or NeighCell RSRQ ≤ PCell RSRQ	Handover	-3	Penalty for poor handover (worse neighbor)
NeighCell RSRP > PCell RSRP and NeighCell RSRQ > PCell RSRQ	Handover	+4	Reward for selecting a better neighboring cell
CINR < 5 after handover	Handover	-1	Extra penalty for poor signal quality after handover
CINR > 20 after handover	Handover	+3	Reward for achieving high signal quality after handover
Action = 0 (No Handover)	No Handover	-	Decision to stay connected to the current cell
CINR > 15 without handover	No Handover	+3	Reward for maintaining good connection quality
CINR ≤ 15 without handover	No Handover	+1	Mild reward for staying connected with

			moderate quality

Table 2: Reward Function

4.3.1.2 Exploration-Exploitation Strategy

Reinforcement learning requires a balance between exploration (trying new actions) and exploitation (leveraging learned actions). Instead of using random action selection for exploration, we incorporated Logistic Regression-based action selection, making the model a hybrid Q-learning approach.

Early-stage learning: Higher exploration rate ensures the agent samples different scenarios.

Exploration decay: Over time, exploration decreases, allowing the agent to exploit learned policies from the Q-table.

Action selection via Logistic Regression: Instead of choosing actions randomly during exploration, we use a Logistic Regression model to predict the optimal action, providing a structured and informed decision-making process.

4.3.2 Q-Table Update and Policy Learning

The Q-learning update rule is applied iteratively:

$$Q(S_t,a_t) \leftarrow Q(S_t,a_t) + \alpha[\, R_t + \beta \max Q(S_{t+1}, a) - Q(s\_t, a\_t)\, Q(S_t,a_t)]$$

where:

α is the learning rate controlling the update step.

β is the discount factor, determining the importance of future rewards.

max $Q(S_{t+1}, a)$ represents the best expected reward from the next state.

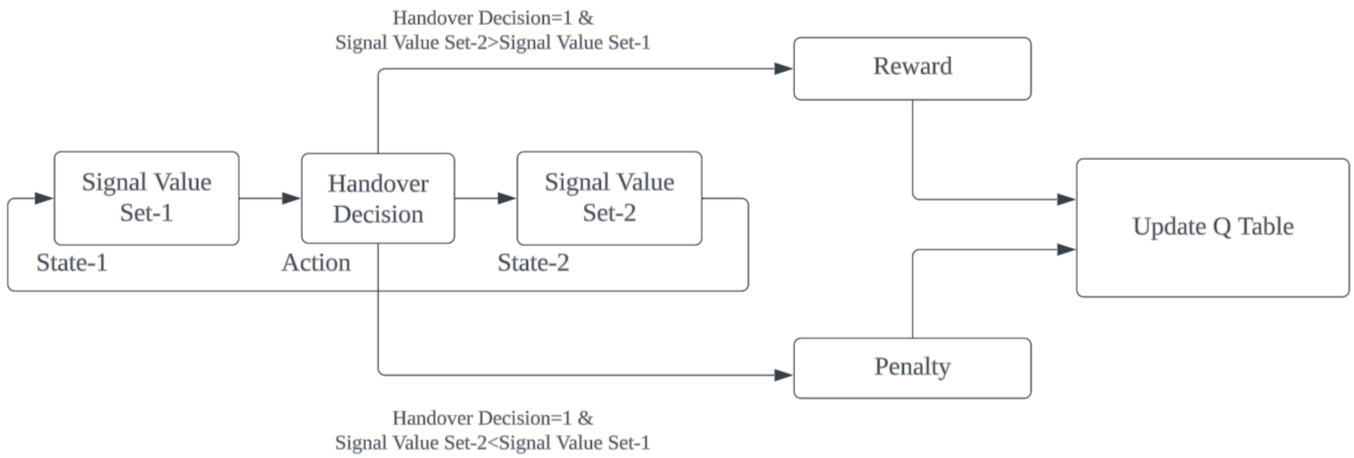

Fig 15: Q learning: Training Phase

By continuously updating the Q-table, the model refines its handover decisions, learning an optimal policy over time.

4.3.3 Evaluation and Handover Optimization

After training, the Q-table was evaluated to assess its impact on handover efficiency. The primary optimization parameter was the number of handovers, which was significantly reduced while ensuring seamless connectivity. This approach enables adaptive, data-driven handover optimization, reducing unnecessary transitions and enhancing network performance.

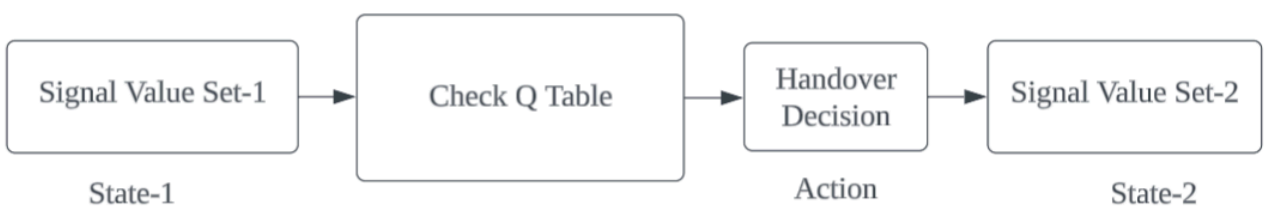

Fig 16: Q learning: Evaluation Phase

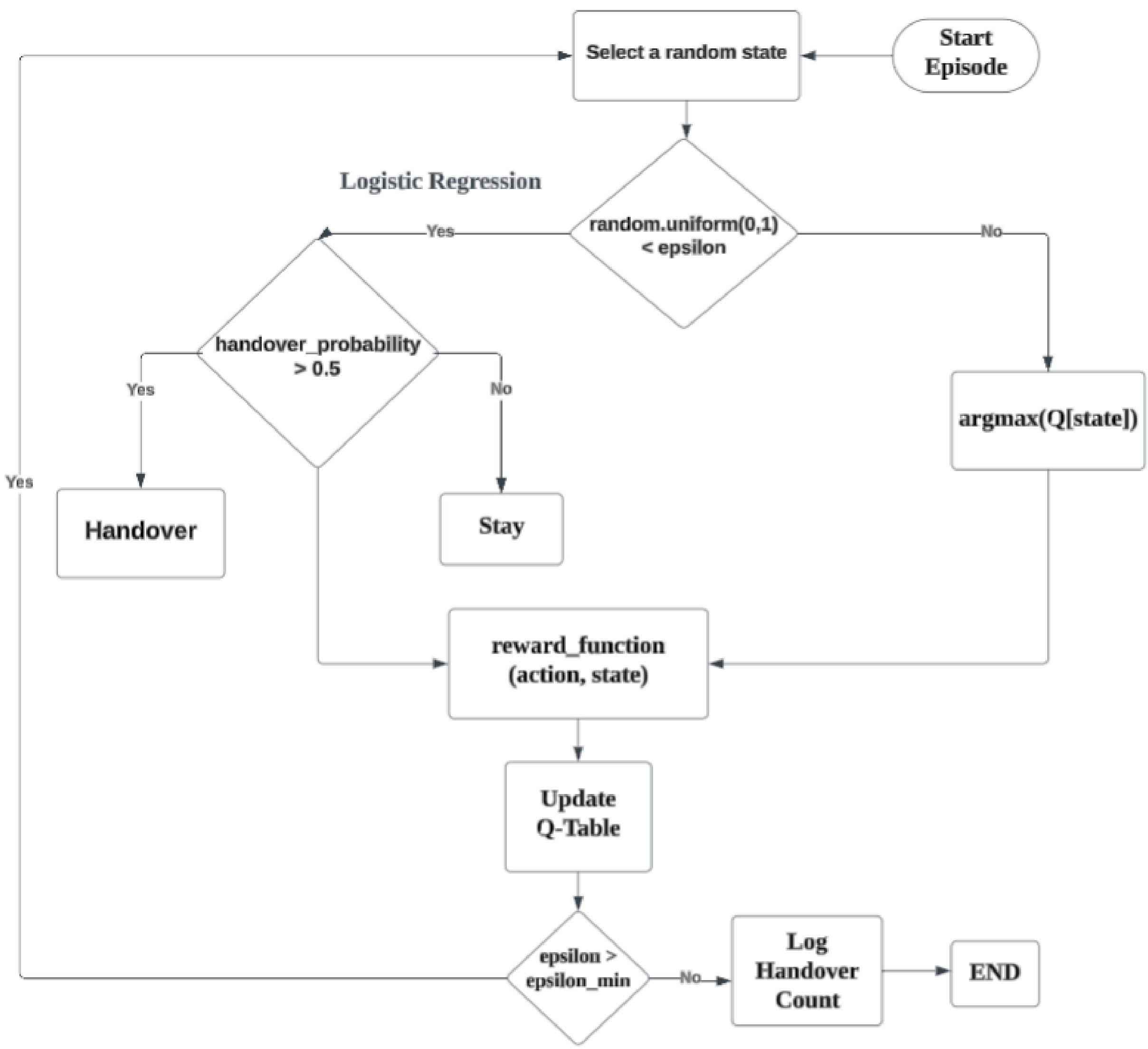

Fig 17: Flow Chart of the proposed algorithm

4.4 Limitations of the Methodology

The proposed methodology optimizes handover decisions by minimizing the number of handovers using a hybrid Q-learning approach. However, it does not account for other critical network performance metrics such as latency, throughput, and packet loss, which are essential for overall Quality of Service (QoS). The optimization framework primarily focuses on reducing unnecessary handovers, but its impact on end-to-end network performance remains unaddressed. Future work can extend the model to incorporate multi-objective optimization, considering additional performance parameters for a more comprehensive handover management strategy.

5. Results and Analysis

This section presents a comprehensive evaluation of the proposed model based on key performance metrics and experimental results. The analysis is structured into three main subsections: Quality Signal Indicator (QSI), handover classification, and handover optimization using hybrid Q-learning. First, the QSI assessment provides insights into signal quality distribution and its correlation with handover events. Next, the handover classification results are examined, comparing model performance against baseline approaches. Finally, the effectiveness of hybrid Q-learning in optimizing handover decisions is evaluated, demonstrating its impact on network performance.

5.1 Experimental result evaluation of QSI

For QSI analysis, we need only 3 features: serving cell RSRP, serving cell RSRQ and CINR. As per the methodology previously described in 4.1 section, first we performed self supervised learning and labeled the unlabeled data. We categorize them into 4 groups: Excellent, Good, Moderate and Poor. Table-2 shows the threshold mapping for the signal quality data.

Category	Threshold Quantile Percentage
Excellent	>90%
Good	>50%
Moderate	>15%
Poor	<15%

Table-3: Threshold Mapping

Figure 18 presents a comparison of the accuracy of the nine models evaluated in this study. Among the classifiers, Logistic Regression and Support Vector Machine (SVM) achieved the highest accuracy, demonstrating superior performance in predicting Quality Signal Indicator (QSI) categories. In contrast, AdaBoost showed the poorest performance, with notably lower accuracy compared to the other models. This comparison highlights the varying effectiveness of the classifiers in handling multiclass classification tasks for signal quality prediction.

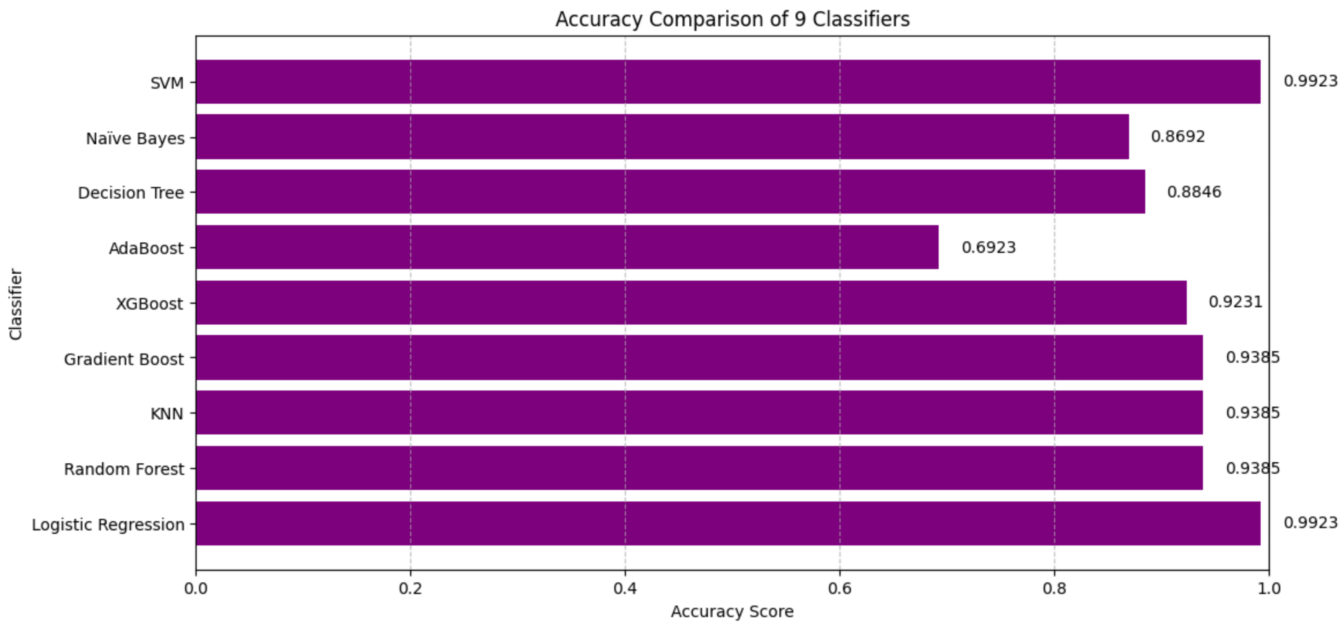

Fig 18 : QSI Model Accuracy

We applied Principal Component Analysis (PCA) to reduce the dimensionality of the dataset while preserving its key variance. Following the PCA transformation, we performed clustering to group similar data points based on their intrinsic patterns. The results were visualized using a scatter plot to illustrate the distribution and separability of the identified clusters. Figure 16 presents the scatter plot, highlighting the clustered categories and their relationships in the transformed feature space.

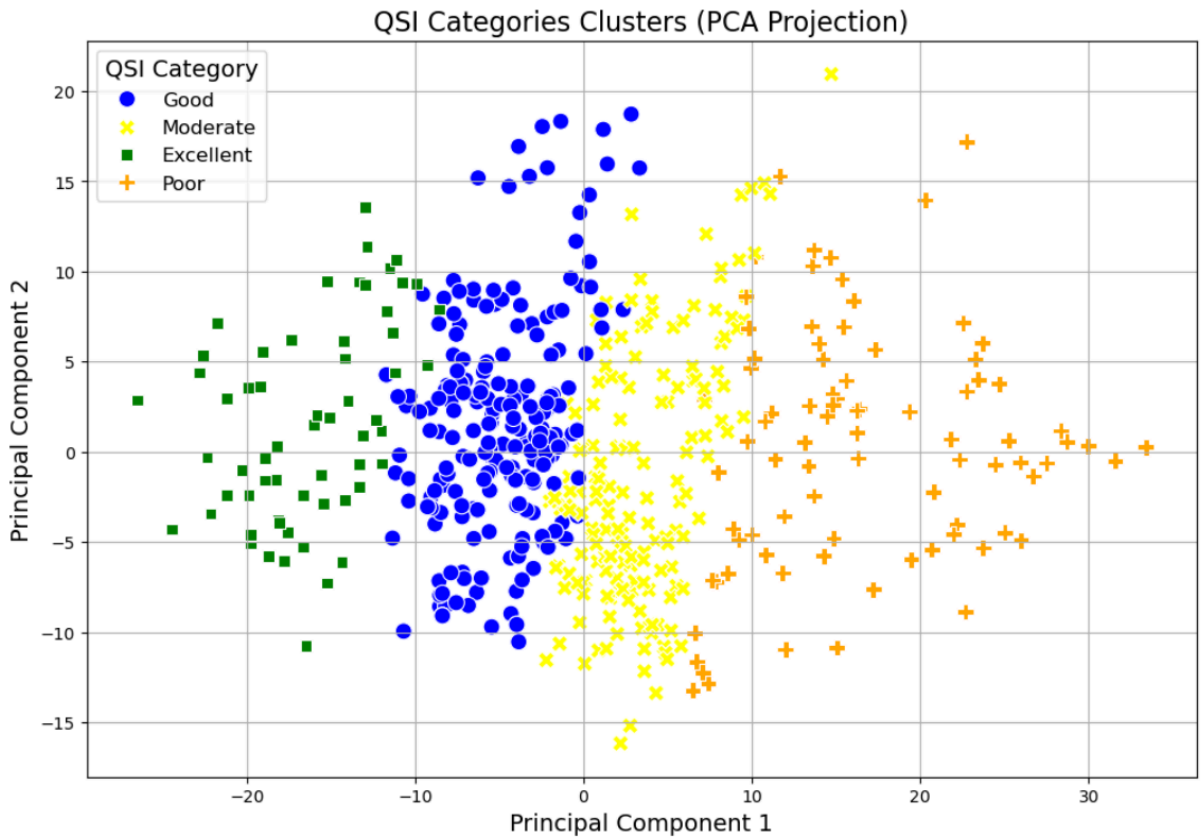

Fig: 19: QSI Clustering using Predefined Labeling

Now we extracted the centroids from this scatter plot of each category. Table-4 shows the PCA axis positions of the centroids.

QSI_Category	PCA-1	PCA-2
Excellent	-16.042605	1.767194
Good	-4.831206	1.473416
Moderate	3.20405	-2.391944
Poor	16.350566	0.374912

Table-4: Centroid Positions

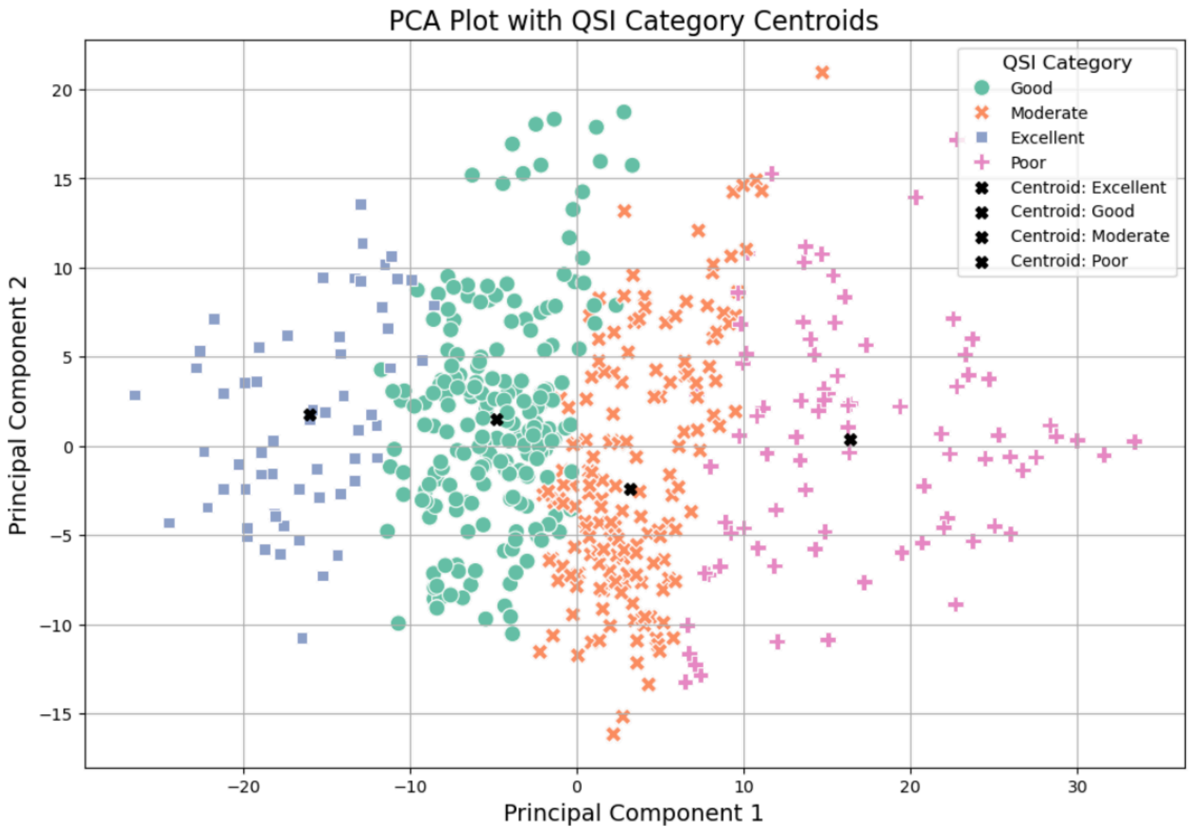

Fig 20: QSI clustering with centroids

Now,we employed **K-Means clustering**, an unsupervised machine learning algorithm, to classify Quality Signal Indicator (QSI) categories based on predefined centroids. The centroids were manually initialized using domain-specific knowledge, representing four distinct QSI categories: **Excellent, Good, Moderate, and Poor**. By specifying these centroids as the initial input, we ensured a guided clustering process that aligns with real-world network conditions. The K-Means algorithm was then executed with **four clusters, a fixed initialization, and a maximum of 200 iterations**, optimizing cluster assignments while maintaining consistency. Finally, the obtained cluster labels were mapped to their corresponding QSI categories, enabling a structured classification of signal quality levels in the dataset.

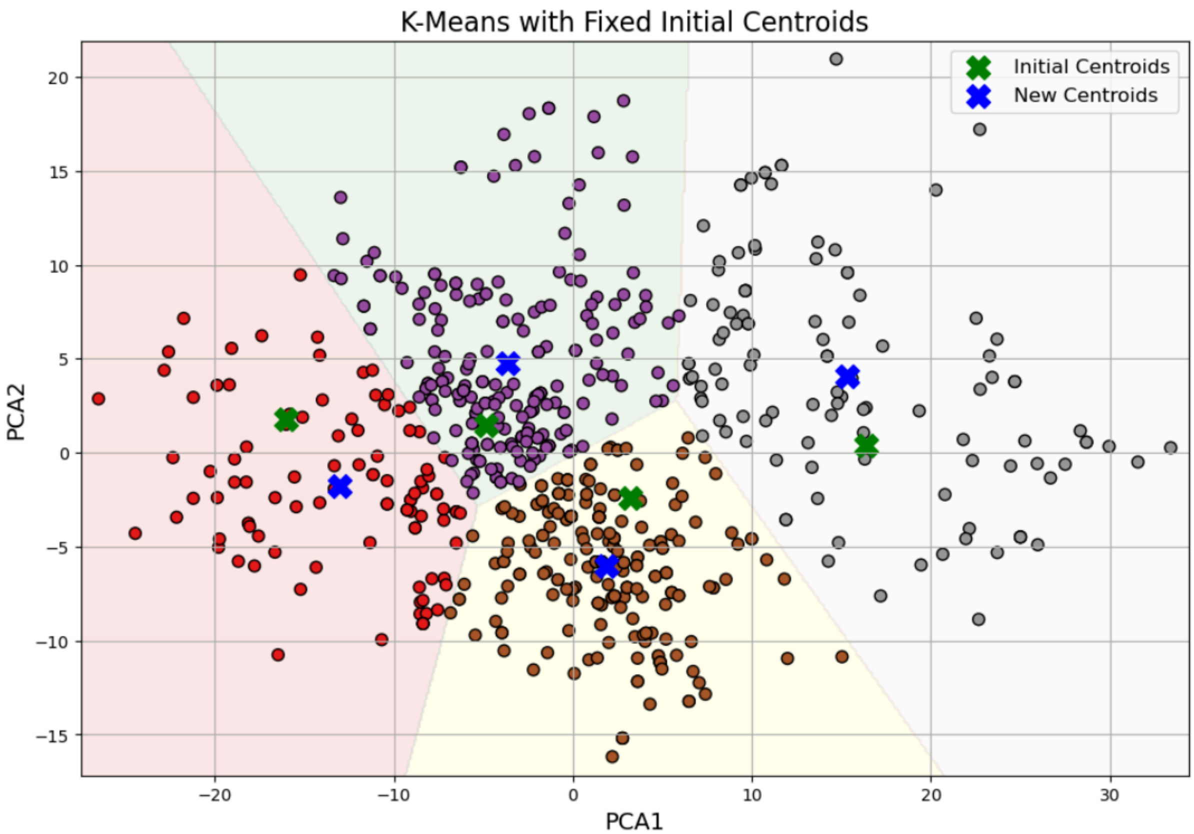

Fig 21: QSI Clustering using K-Means algorithm

5.2 Experimental result evaluation of Handover Prediction:

Evaluation Metrics

Evaluation metrics are quantitative measures used to assess the performance of a machine learning model. These metrics help determine how well a model makes predictions and whether it meets the desired objectives. The choice of evaluation metric depends on the type of problem being solved, such as classification, regression, or clustering.

Precision

Precision measures how many of the instances predicted as positive are actually positive. It is calculated as Precision = TP / (TP + FP), where TP (True Positives) are correctly predicted positive cases, and FP (False Positives) are incorrectly predicted positives. A high precision means fewer false positives, making it crucial in applications where false alarms are costly, such as medical diagnoses or fraud detection.

Recall

Recall, also known as sensitivity, measures how well the model captures actual positive cases. It is defined as Recall = TP / (TP + FN), where FN (False Negatives) are actual positives that were incorrectly classified as negative. A high recall ensures that most positive cases are identified, which is important in applications like disease screening, where missing a positive case can have severe consequences.

F1-Score

F1-score is the harmonic mean of precision and recall, balancing both metrics. It is calculated as F1-score = 2 × (Precision × Recall) / (Precision + Recall). This metric is particularly useful when there is an imbalance in the dataset, as it provides a single value that accounts for both false positives and false negatives. A high F1-score indicates that the model maintains a good trade-off between precision and recall.

Accuracy

Accuracy is a commonly used metric that measures the overall correctness of the model's predictions. It is defined as Accuracy = (TP + TN) / (TP + TN + FP + FN), where TN (True Negatives) are correctly predicted negative cases. While accuracy is useful for balanced datasets, it can be misleading in imbalanced datasets where one class dominates.

To evaluate the performance of the handover prediction model, we employed ten different classification algorithms, ensuring a comprehensive assessment of predictive accuracy. The proposed model architecture, as detailed in Section 4.2, was designed to enhance classification efficiency by incorporating imbalanced dataset handling techniques. The following figures illustrate a comparative analysis of performance metrics, including accuracy, precision, recall, and F1-score, between the baseline model and the proposed model. This evaluation provides valuable insights into the effectiveness of the proposed approach in improving handover prediction outcomes.

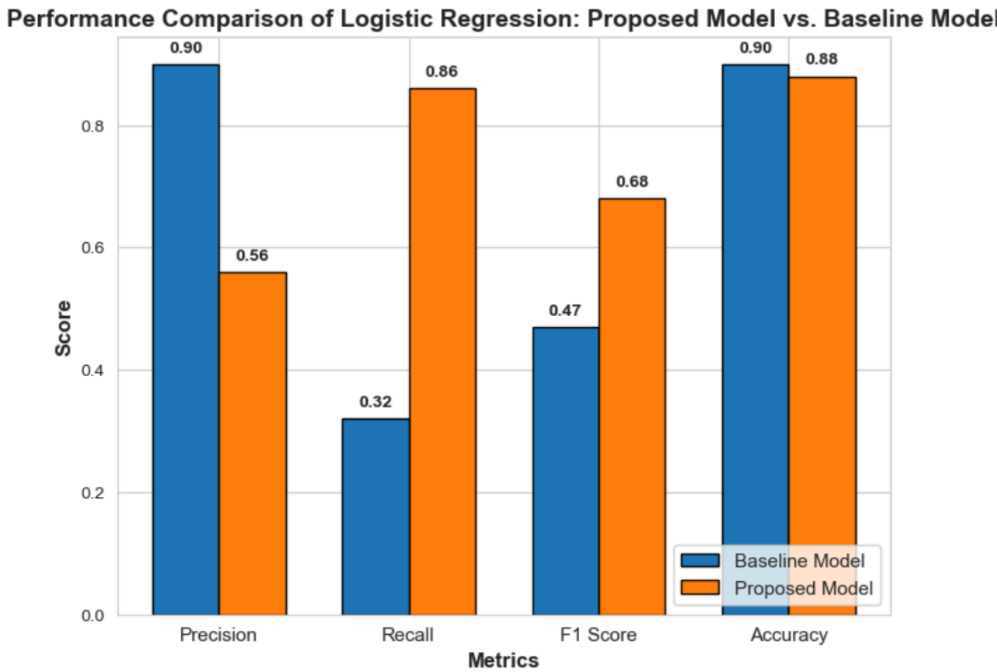

Fig 22: Logistic regression

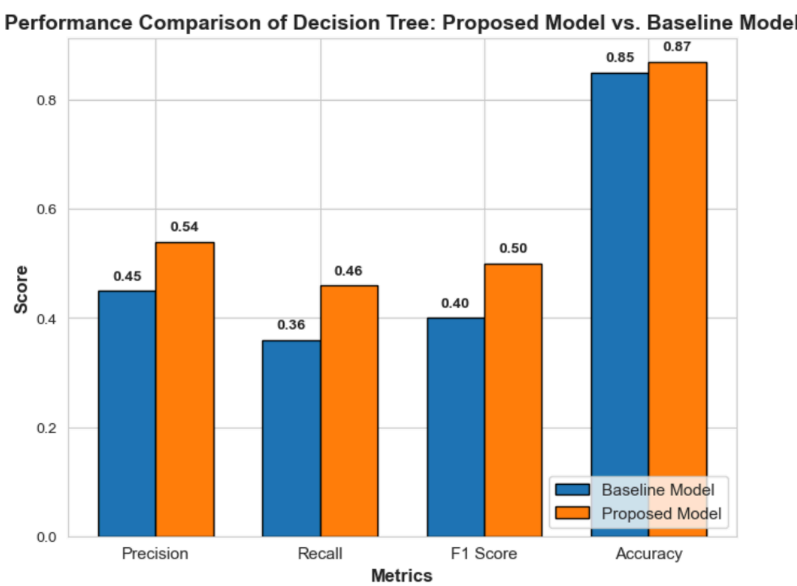

Fig 23: Decision tree

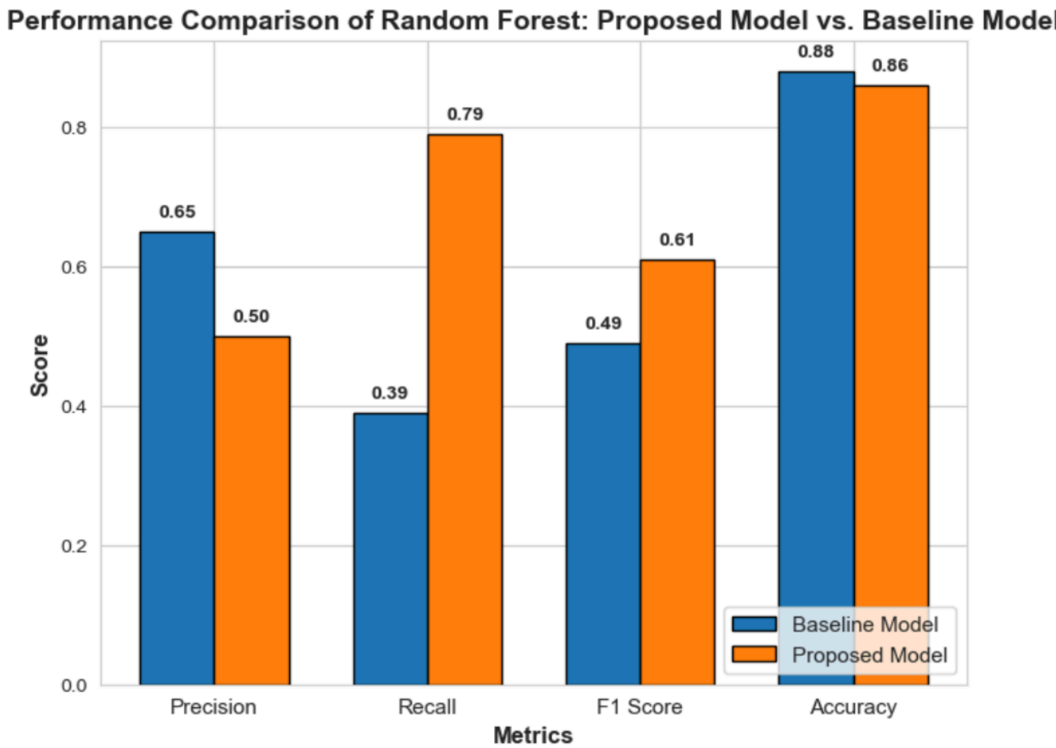

Fig 24: Random Forest

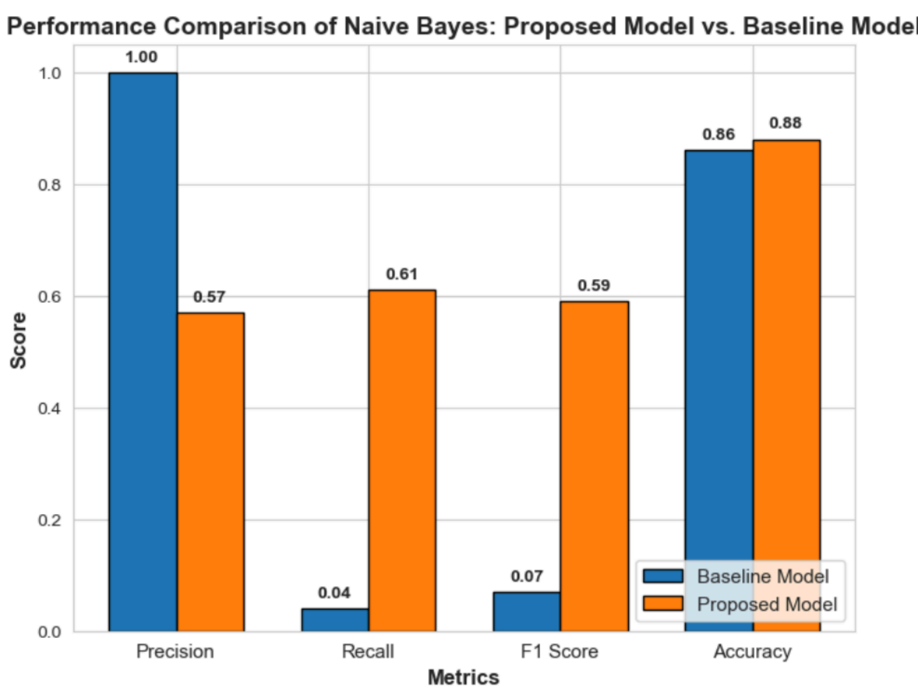

Fig 25: Naive Bayes.

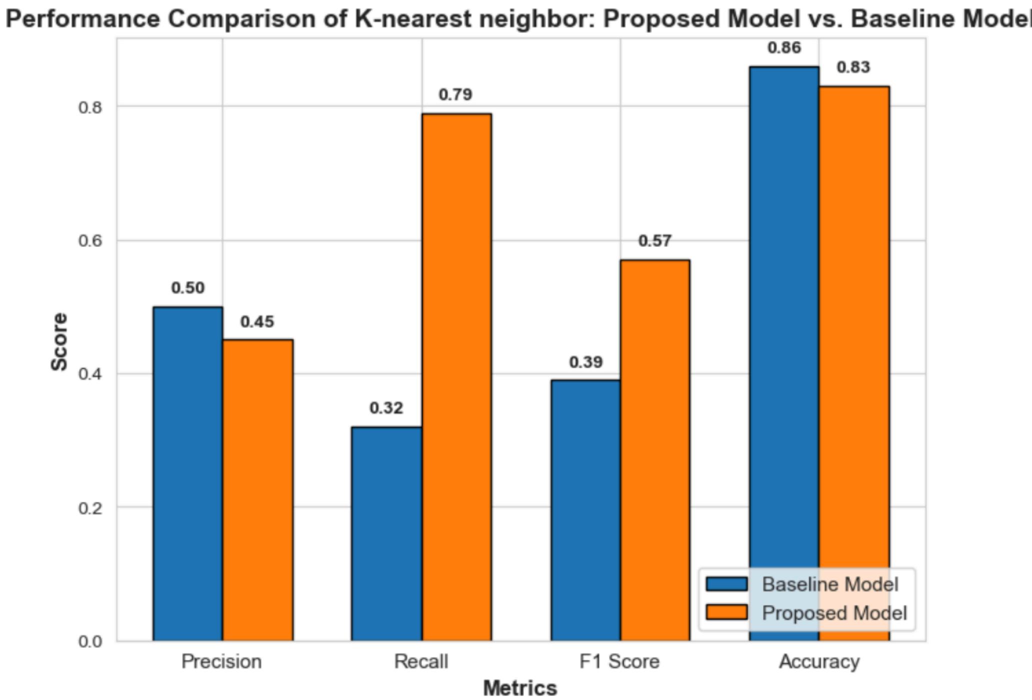

Fig 26: K-nearest neighbor

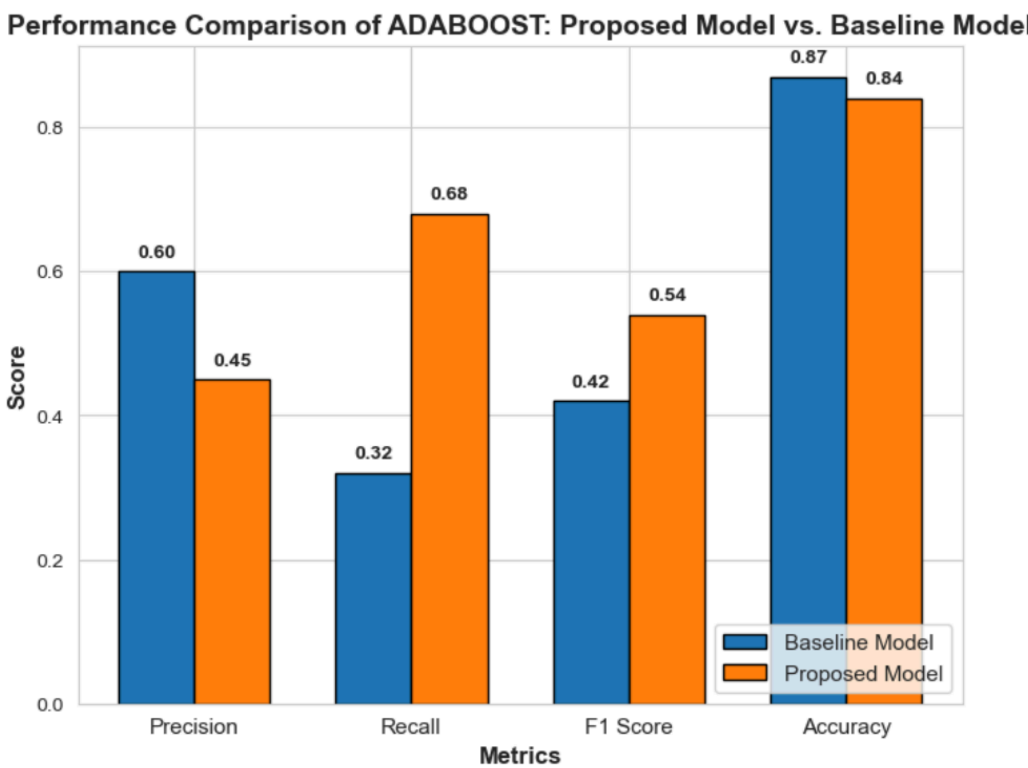

Fig 27: ADAboost

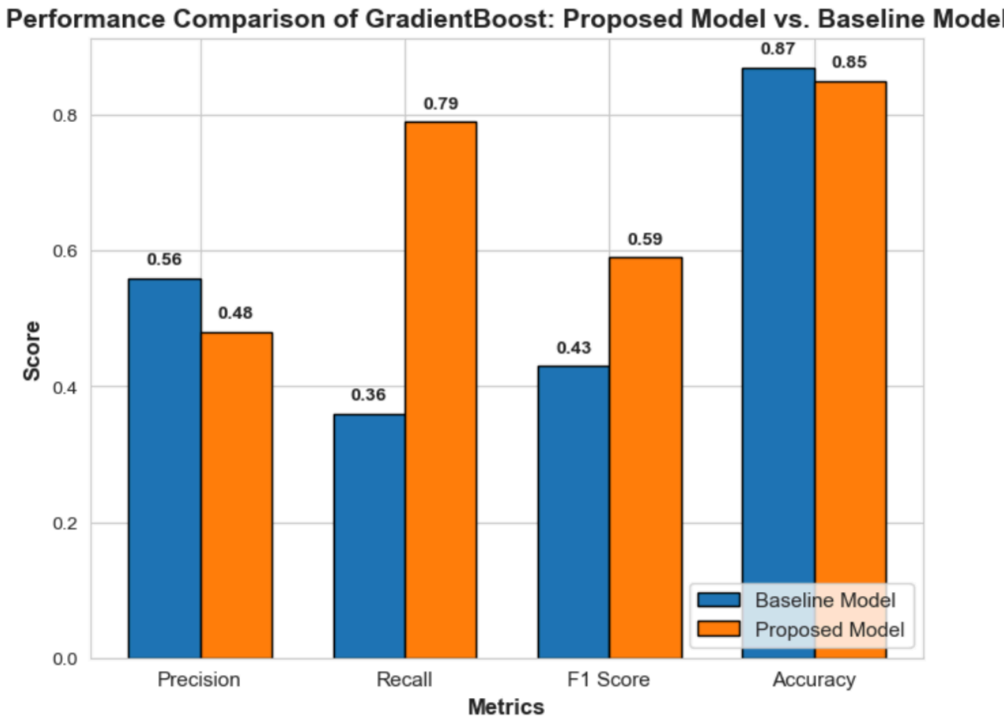

Fig 28: Gradient Boost

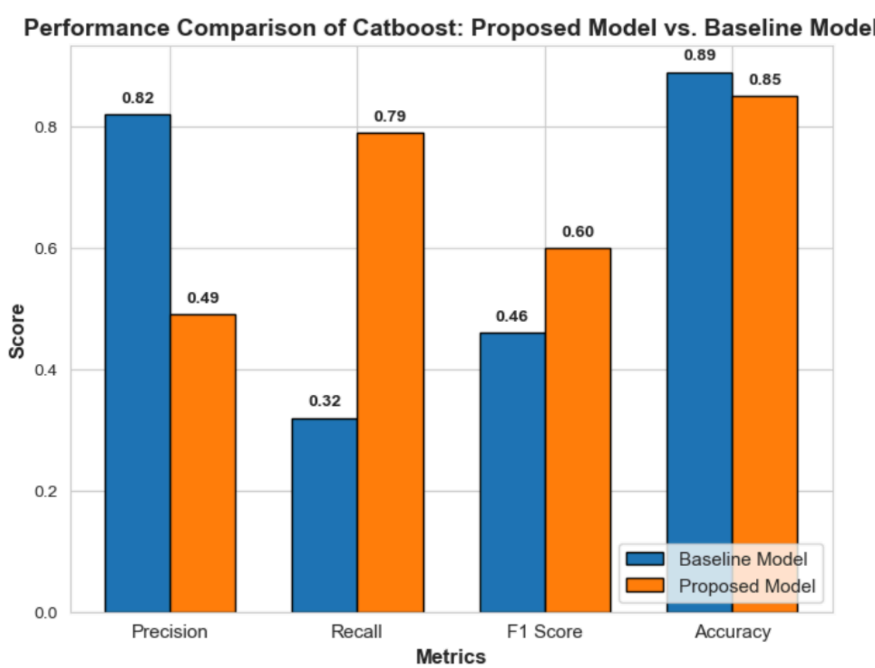

Fig 29: Catboost

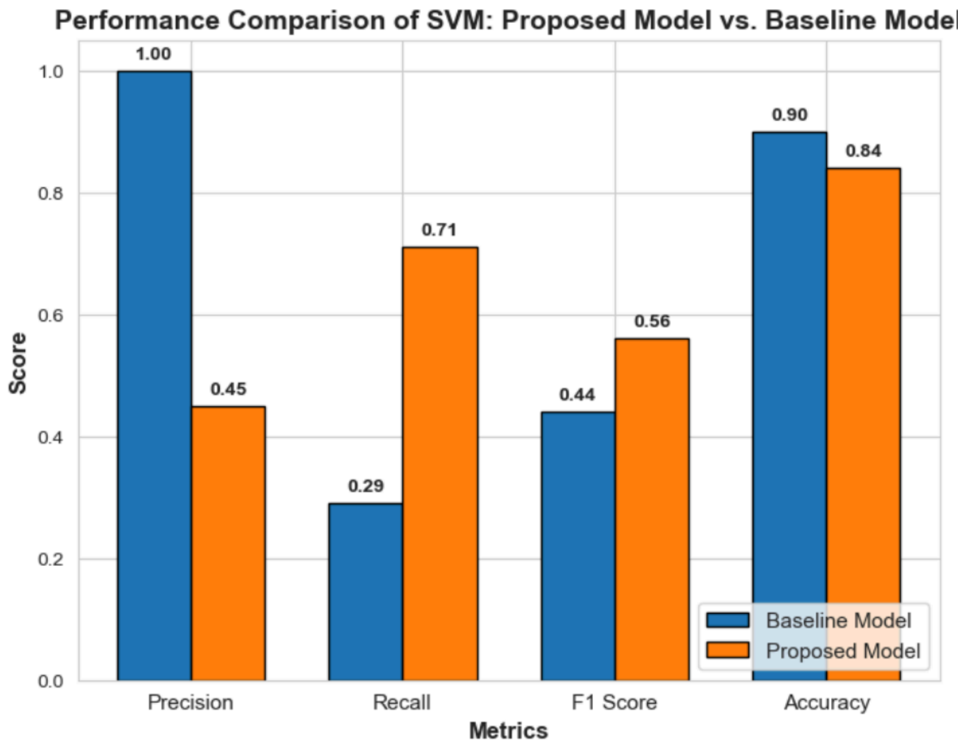

Fig 30: SVM

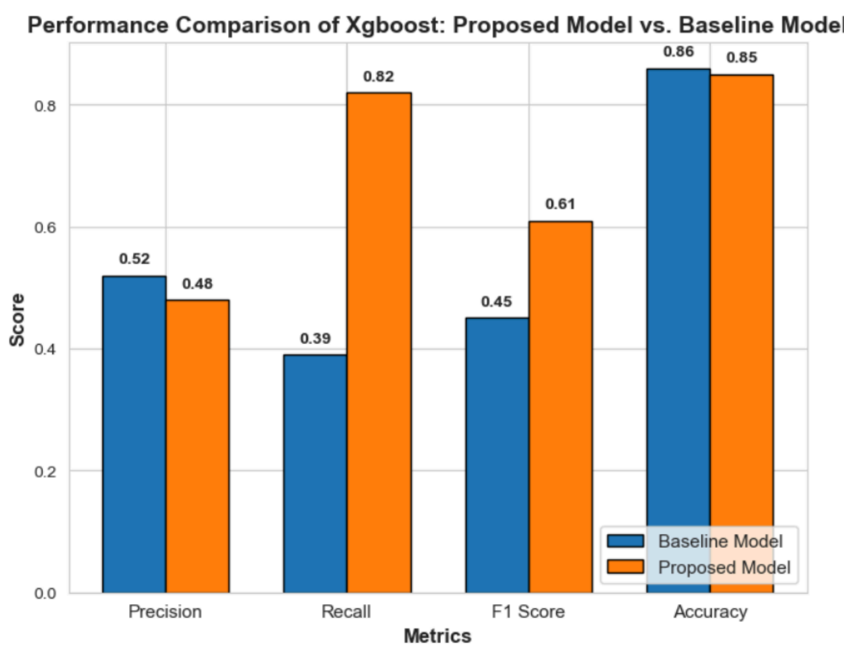

Fig 31: XGboost

The confusion matrices of each classifier are shown for better understanding in the following figures.

Logistic Regression:

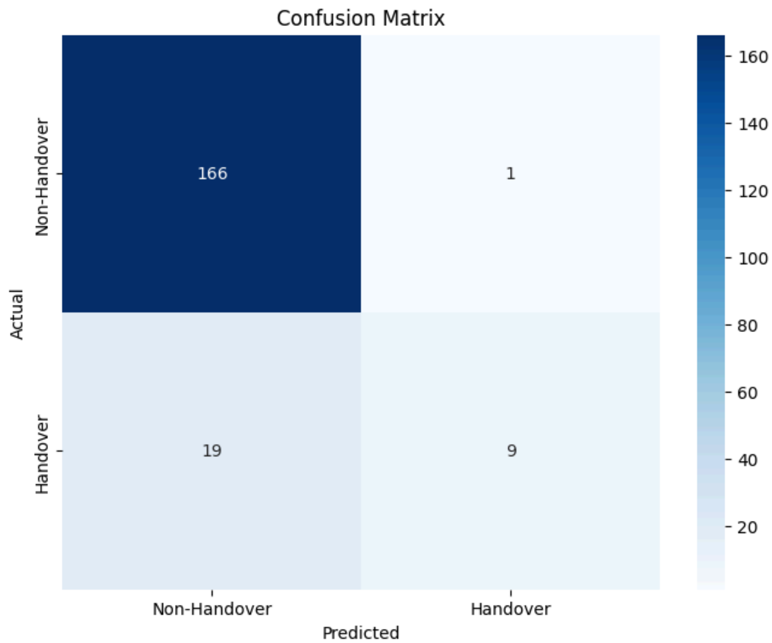

Fig 32: Baseline Model

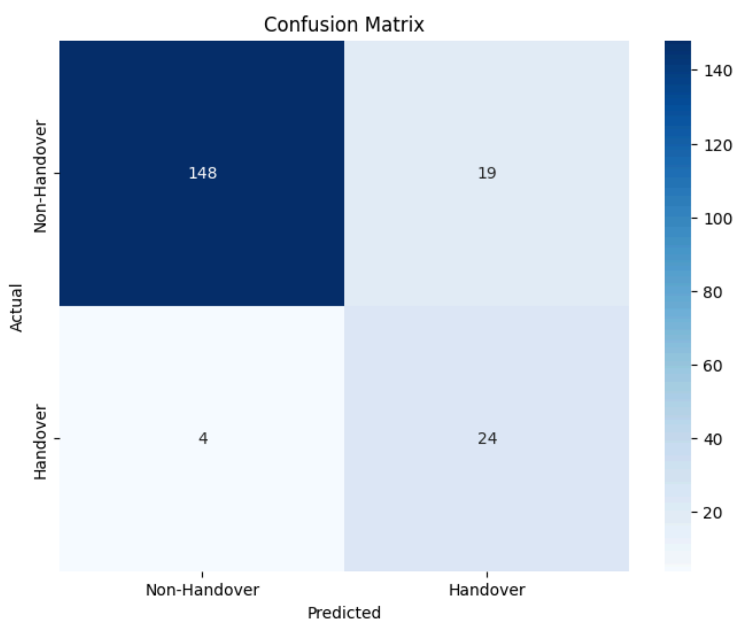

Fig 33: Proposed Model

Decision Tree:

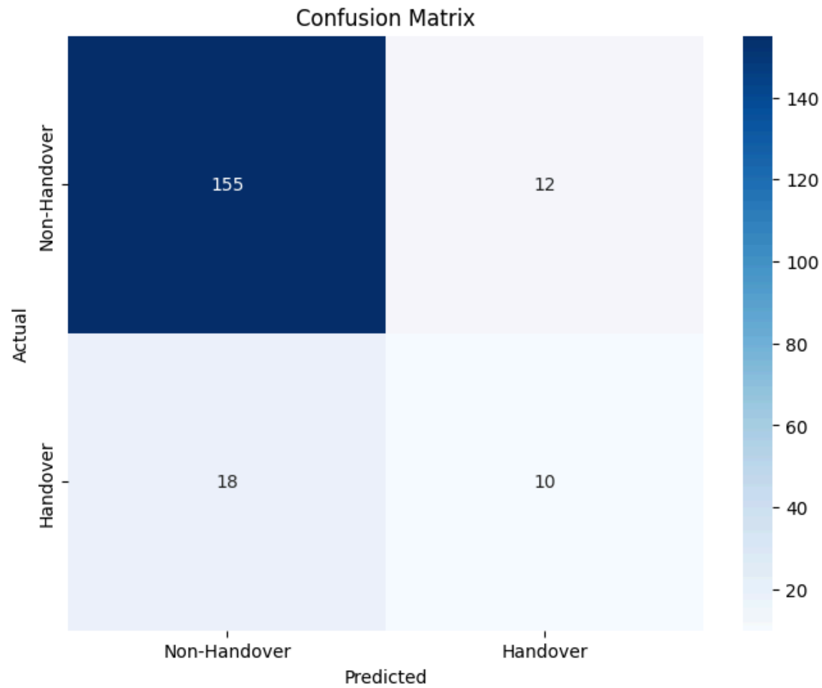

Fig 34: Baseline Model

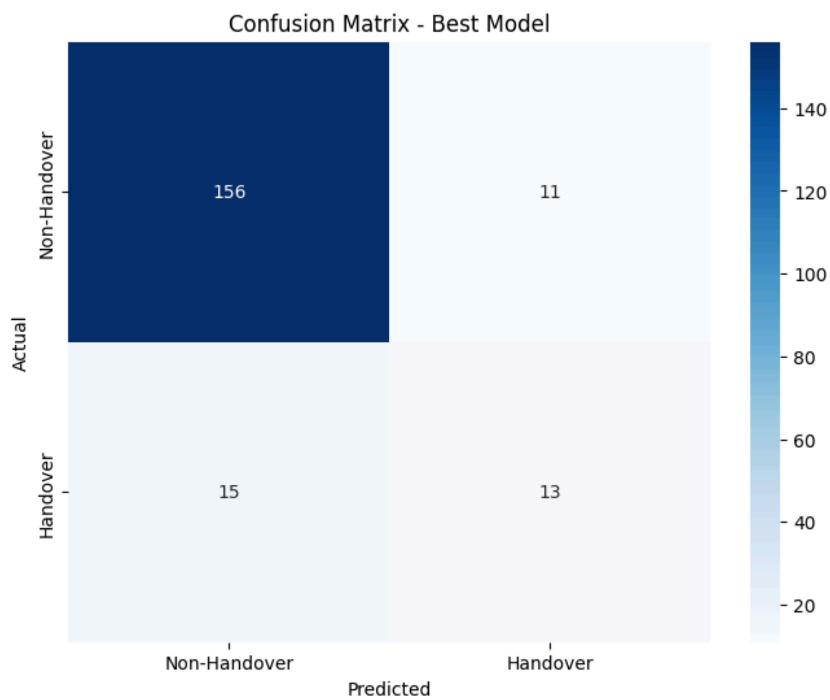

Fig 35: Proposed Model

Random Forest:

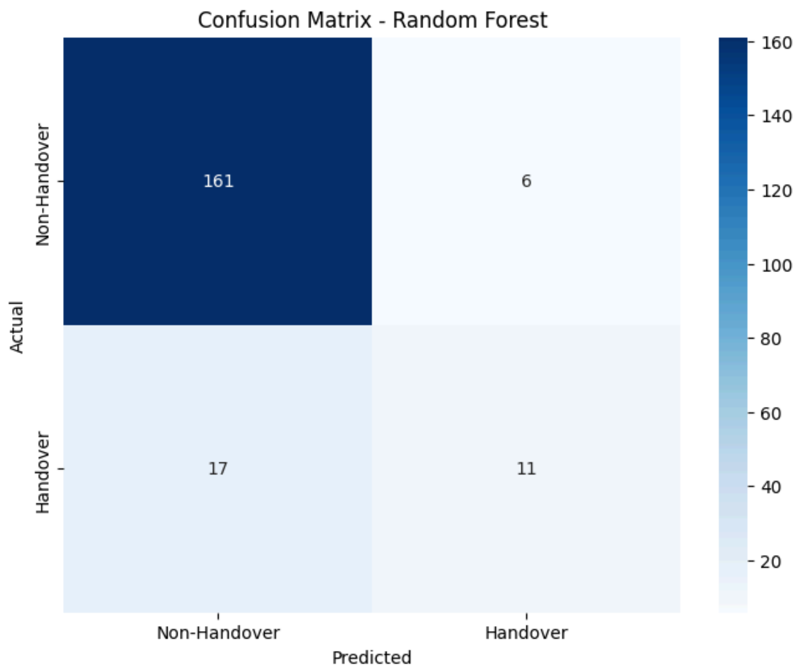

Fig 36: Baseline Model

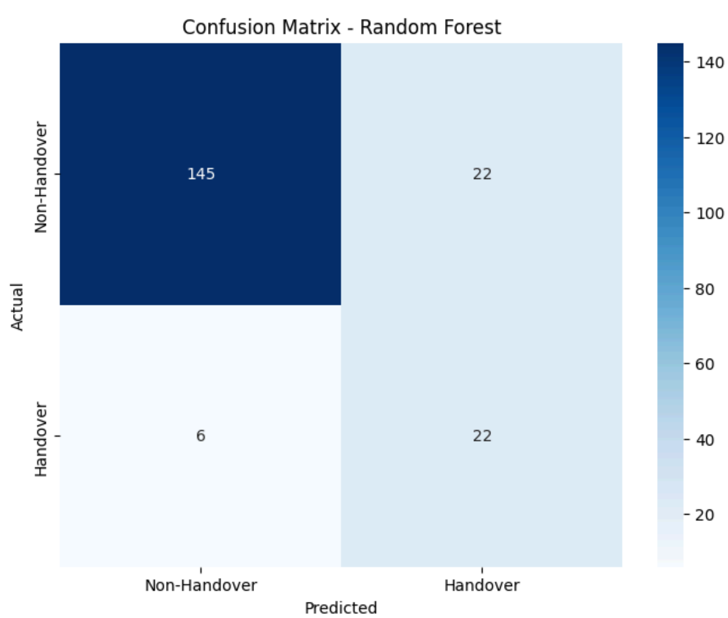

Fig 37: Proposed Model

KNN:

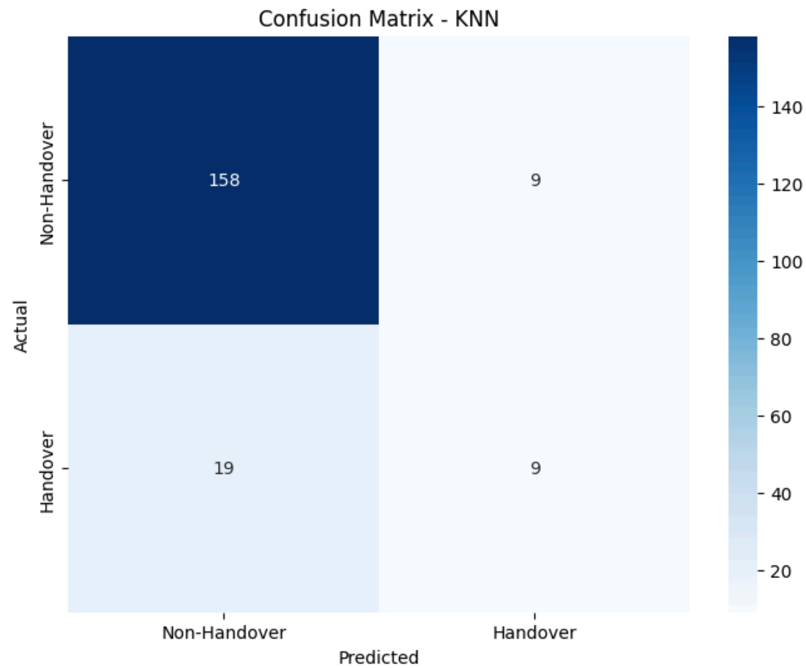

Fig 38: Baseline Model

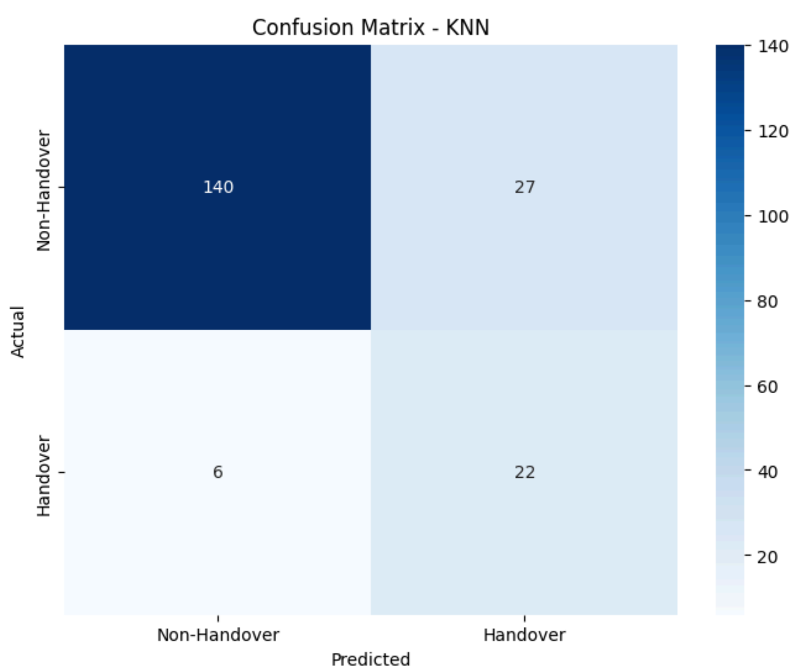

Fig 39: Proposed Model

SVM:

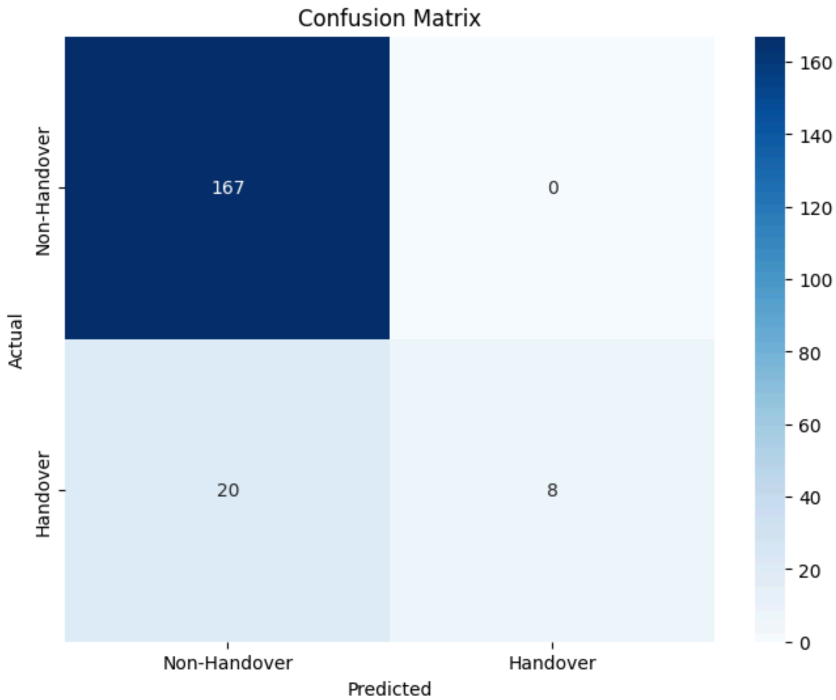

Fig 40: Baseline Model

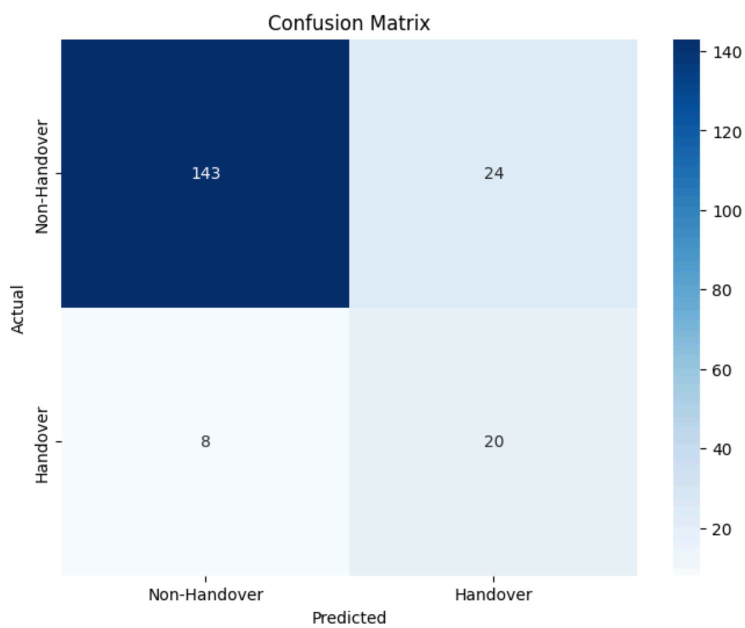

Fig 41: Proposed Model

Naive Bayes:

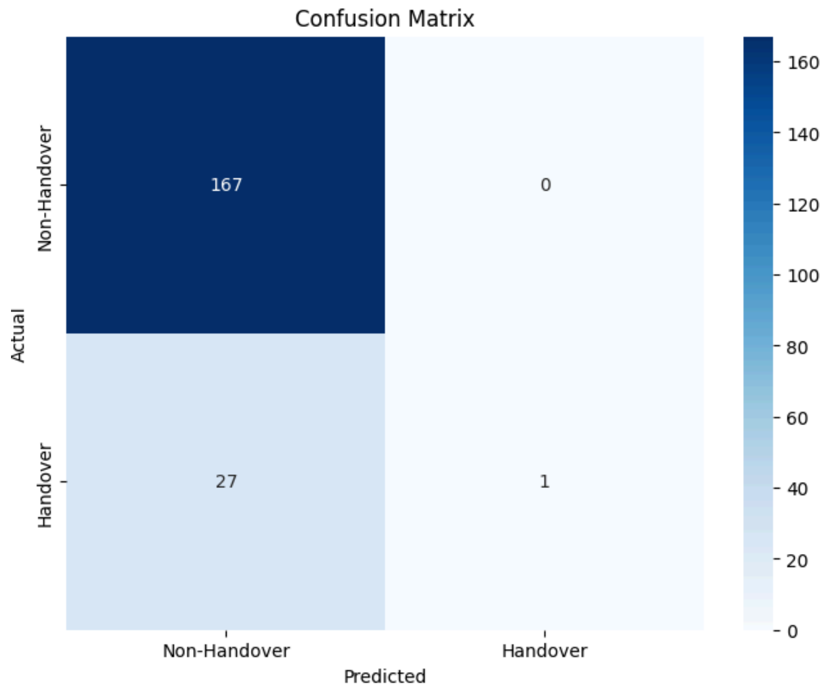

Fig 42: Baseline Model

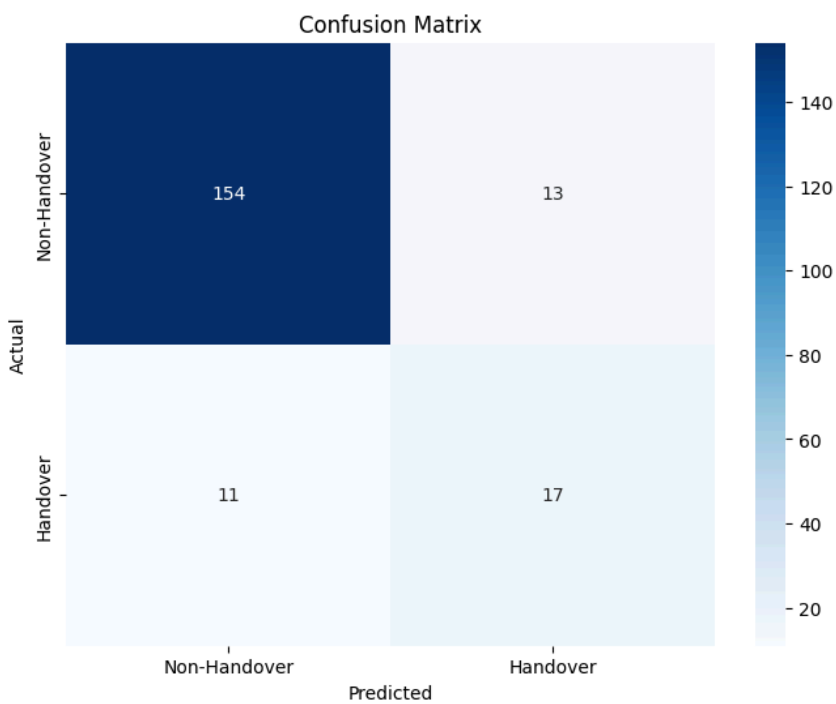

Fig 43: Proposed Model

Adaboost:

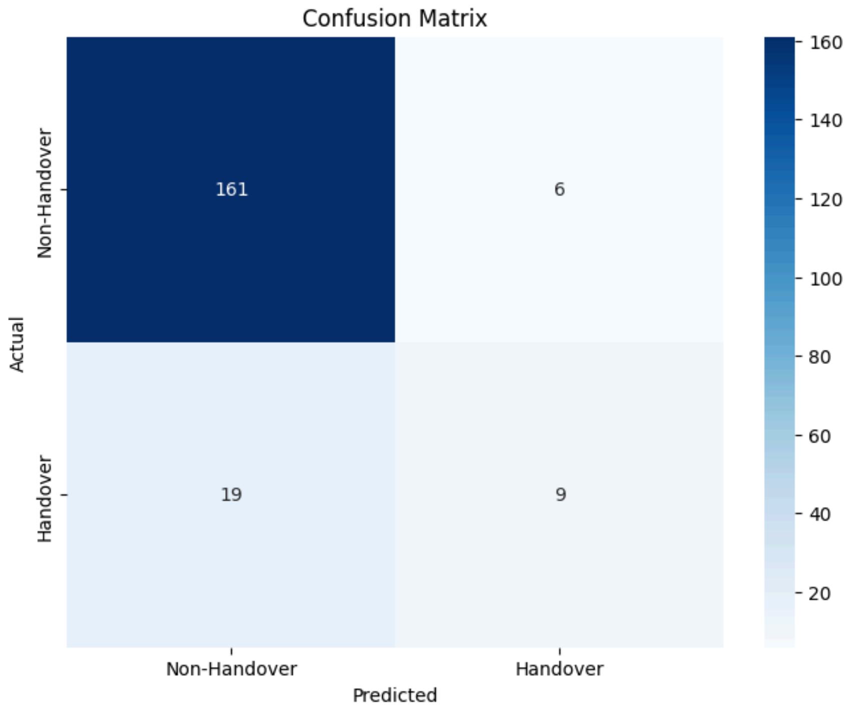

Fig 44: Baseline Model

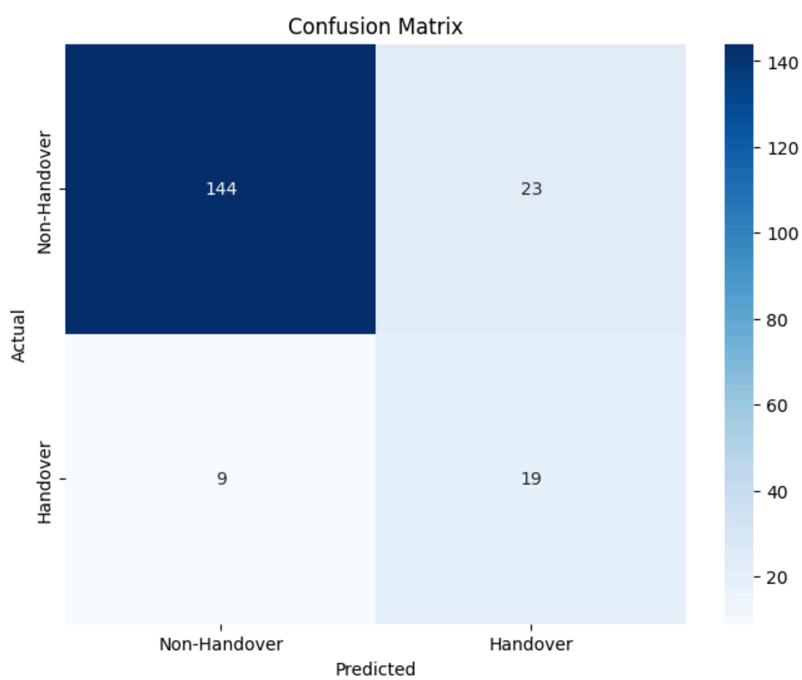

Fig 45: Proposed Model

Catboost:

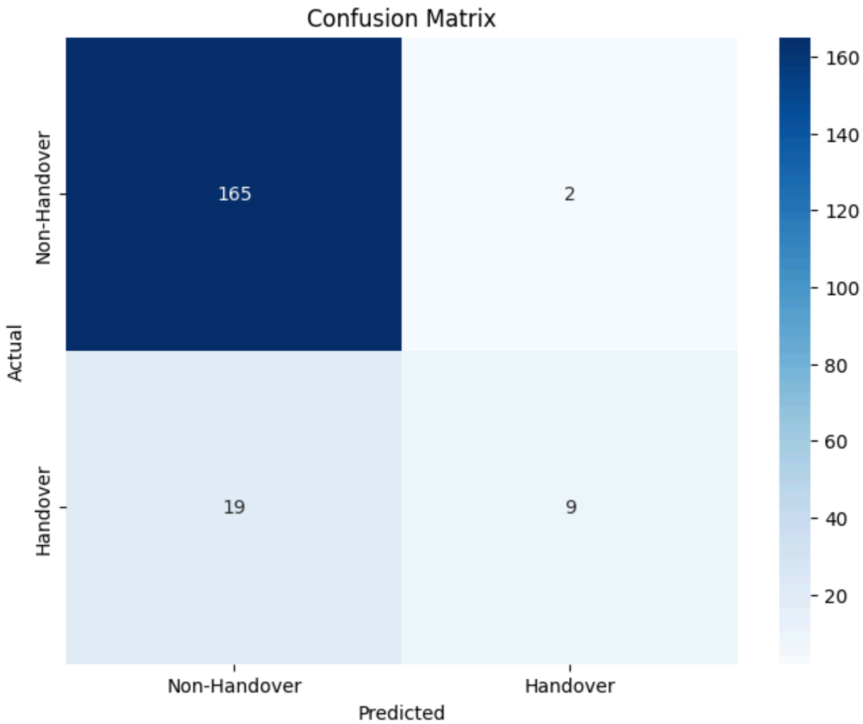

Fig 46: Baseline Model

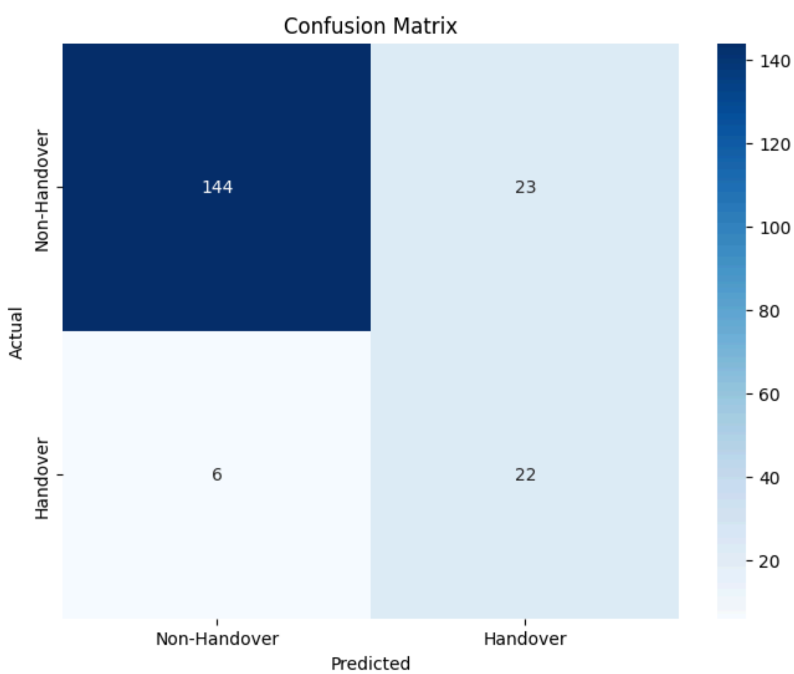

Fig 47: Proposed Model

Xgboost:

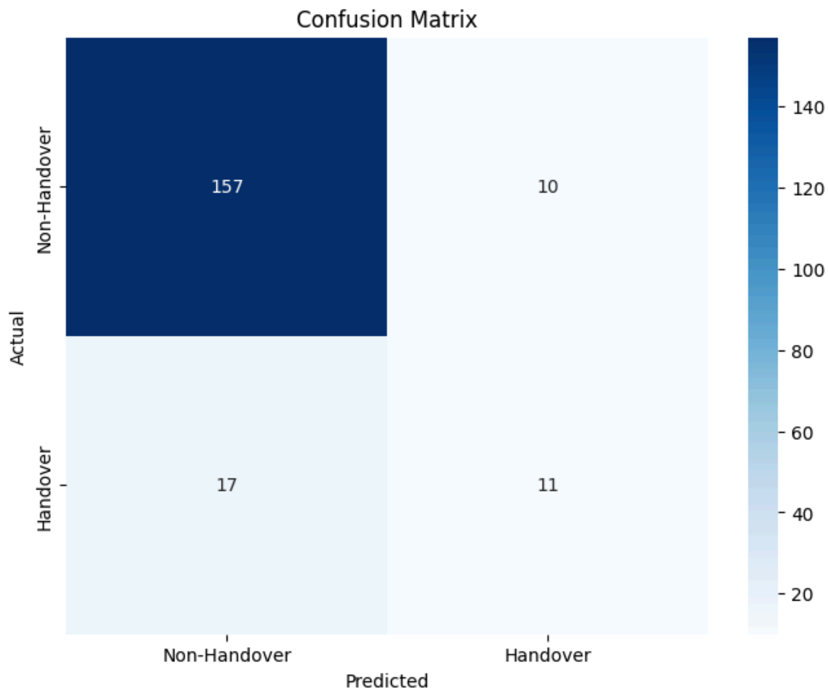

Fig 48: Baseline Model

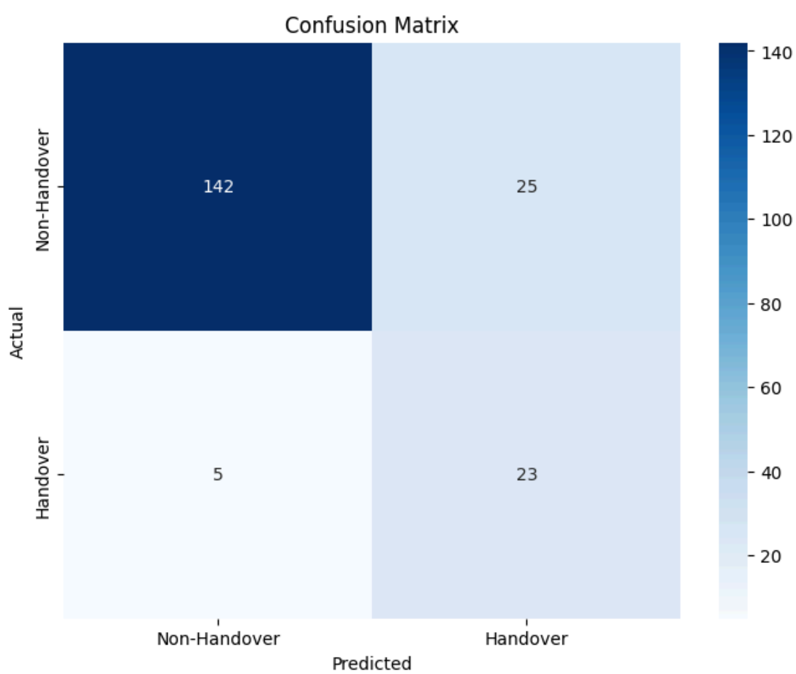

Fig 49: Proposed Model

Gradiebtboost:

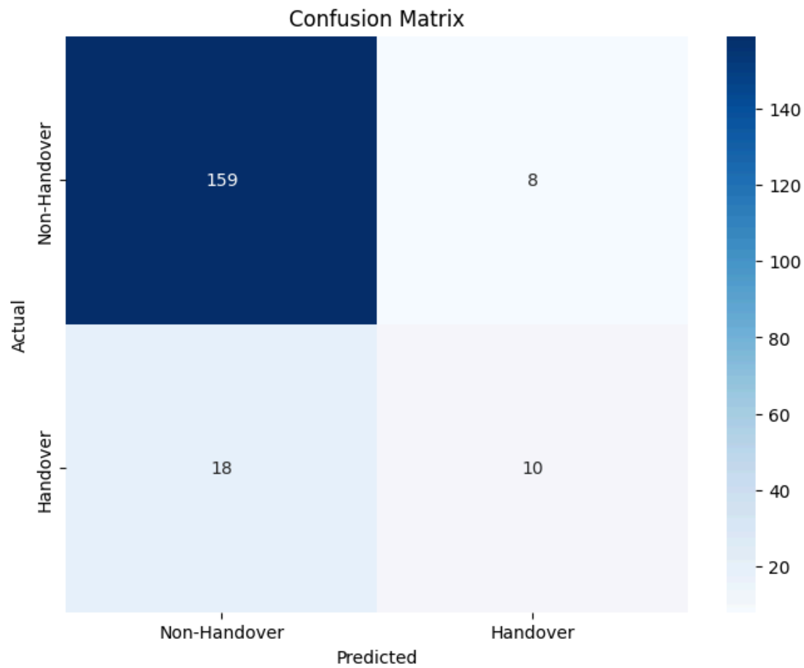

Fig 50: Baseline Model

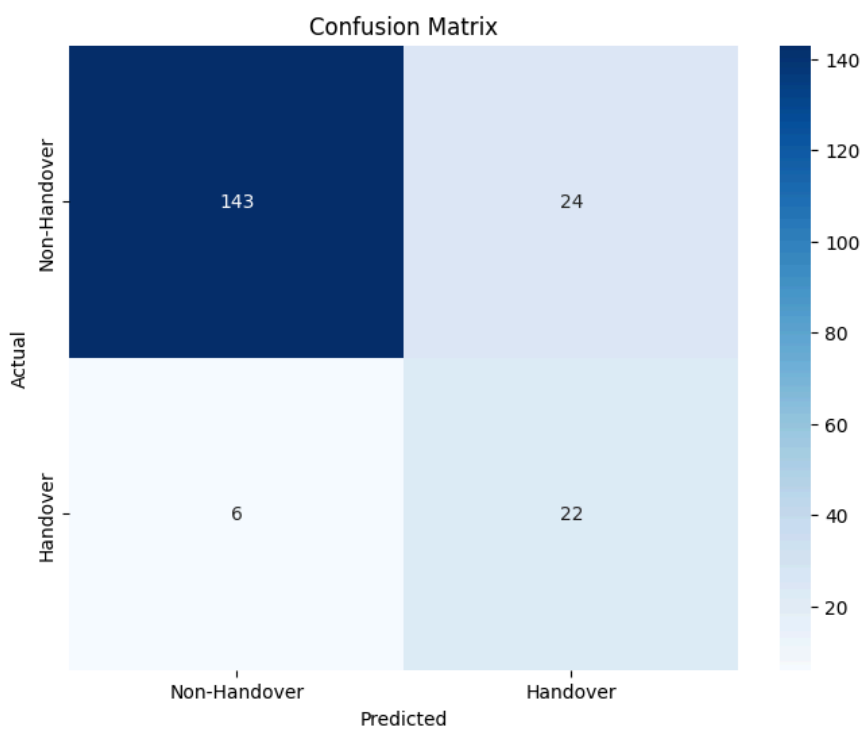

Fig 51: Proposed Model

The proposed model demonstrated a significant improvement over the baseline models in terms of Recall and F1-score, indicating a higher capability of capturing actual positive instances while maintaining a balanced trade-off between precision and recall. However, a slight reduction in Precision was observed, suggesting an increase in false positives. This trade-off was a deliberate consideration in model optimization, prioritizing higher recall to minimize false negatives, which is critical in our application domain. The rationale behind this decision and potential strategies to mitigate the precision drop will be discussed in detail in the subsequent section.

5.3 Experimental result evaluation of Handover Optimization:

As outlined in Section 4.3, our primary objective is to minimize unnecessary handovers to enhance the signal quality of a moving User Equipment (UE). To achieve this, we propose a hybrid learning framework that integrates supervised learning with Q-learning, leveraging a reward-based mechanism to optimize handover decisions. The model is designed to refine classifier-based predictions by strategically mitigating redundant handovers, ensuring that only necessary transitions occur between serving and neighboring cells.

The reward function plays a critical role in the reinforcement learning (RL) component of our model. It systematically penalizes handover decisions that do not yield improved signal quality while rewarding actions that maintain or enhance network performance. By incorporating parameters such as Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), and Carrier-to-Interference-plus-Noise Ratio (CINR) into the reward function, the Q-learning agent learns to differentiate between beneficial and detrimental handover events. This targeted reinforcement mechanism enables the model to dynamically adjust handover thresholds and prioritize network stability over frequent cell transitions.

In our dataset, we initially observed 97 handover instances across 647 data points. During the training phase, our model recorded 276 handover events, allowing the Q-learning agent to iteratively refine its decision-making policy. Subsequently, in the evaluation phase—where the trained Q-table was applied across the entire dataset—the number of handovers was reduced to 71 instances, indicating that the remaining handovers were identified as unnecessary and successfully suppressed. This reduction demonstrates the model's effectiveness in preventing unnecessary handovers while preserving seamless connectivity.

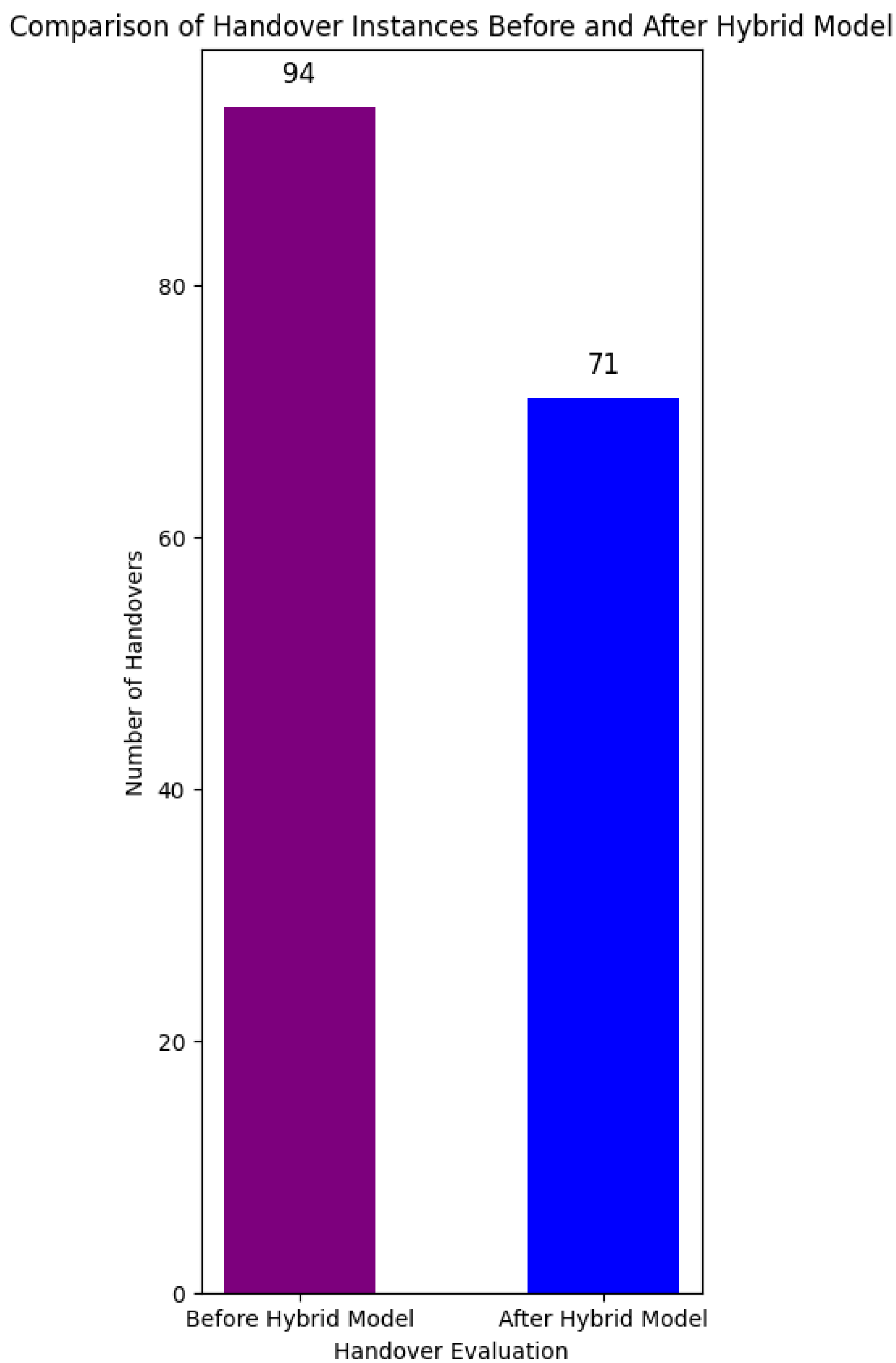

Fig 52: Number of Handover Before and After applying our model

Minimizing redundant handovers is crucial in wireless communication systems, particularly in **LTE and 5G networks**, where frequent transitions between cells can lead to increased signaling overhead, network congestion, and degraded Quality of Service (QoS). Excessive handovers also contribute to **higher latency, packet loss, and energy consumption**, negatively impacting both network efficiency and user experience. By intelligently regulating handover events, our proposed model enhances overall network performance, ensuring improved **signal stability, lower handover failure rates, and optimized resource utilization**. This adaptive framework can be further extended to next-generation cellular networks, facilitating intelligent mobility management in real-world deployments.

6. Discussion and Insights

6.1 Critical Evaluation on QSI

In this study, we employed K-Means clustering to categorize Quality Signal Indicator (QSI) data, using manually specified initial centroids based on predefined QSI categories. Unlike supervised classification, where labels guide the learning process, K-Means operates unsupervised, relying solely on feature-space distances and iterative centroid optimization. While the clustering generally followed the predefined QSI categories, discrepancies arose due to the algorithm's independent reassignment of data points based on proximity to centroids. For instance, some data initially categorized as "Good" were reassigned to "Excellent" in K-Means, demonstrating its data-driven approach. To assess the alignment between the predefined categories and the K-Means results, we calculated category-wise percentage matches, providing insights into the effectiveness of centroid-based initialization in guiding the unsupervised learning process.

Category	Percentage Matches
Excellent	76.92%
Good	65.52%
Moderate	71.43%
Poor	80.41%

Table-5: Performance matching of supervised and unsupervised learning

6.2 Critical Evaluation on Handover Prediction Model

Cost Sensitivity

The underlying cost sensitivity connected to various classes is another element providing unbalanced data. Misclassifying one class can have more negative effects than misclassifying another in several situations. In credit card fraud detection, for instance, the financial institution and the client may suffer large losses in value if a fraudulent transaction is not detected. Misclassifying fraudulent activities therefore comes at a higher cost than misclassifying lawful transactions.

In the context of our study, the dataset exhibits a significant class imbalance, with a notably lower number of handover instances compared to non-handover instances. This imbalance poses a challenge for conventional classification metrics such as accuracy and precision, as these metrics may not provide a comprehensive assessment of model performance under such conditions. Specifically, a classifier that predicts the majority class (non-handover) more frequently can achieve a deceptively high accuracy, even if it fails to correctly identify actual handover events. Similarly, a high precision does not necessarily indicate a well-performing model in this scenario, as it focuses on minimizing false positives while potentially disregarding false negatives.

In our case, recall emerges as a more critical metric, as it quantifies the ability of our model to correctly identify actual handover instances (true positives) while minimizing false negatives. A false negative in this context—where an actual handover event is misclassified as a non-handover—can be costly, leading to degraded network performance, increased packet loss, and potential call drops. Given the high cost associated with undetected handovers, maximizing recall ensures that the model prioritizes detecting handover instances, even at the expense of some false positives.

Moreover, to balance precision and recall, we incorporate the F1-score, which provides a harmonic mean of both metrics. The F1-score is particularly useful in imbalanced classification problems, as it accounts for both false positives and false negatives, ensuring a more holistic evaluation. In our study, despite a slight reduction in precision, a significant improvement in recall and F1-score indicates that the model is effectively identifying handover instances while maintaining an overall balanced performance.

Considering the severe consequences of misclassifying handovers, a cost-sensitive approach is necessary, where false negatives (missed handovers) are penalized more heavily than false positives (unnecessary handovers). Our model is designed with this objective in mind, ensuring that recall remains the dominant metric, as detecting handovers at any cost is the priority to maintain optimal network quality and user experience.

6.3 Critical Evaluation on Handover Optimization

Our Hybrid Q-Learning Model is designed to optimize handover decisions in wireless networks by intelligently reducing unnecessary handovers while ensuring seamless connectivity. This is particularly useful in high-mobility scenarios such as urban cellular networks, where frequent handovers can degrade Quality of Service (QoS).

The model utilizes a user-defined reward function to guide learning, where penalties discourage poor handover decisions (e.g., switching to a weaker cell) and rewards incentivize optimal handovers (e.g., moving to a stronger signal cell with higher CINR, RSRP, and RSRQ). This function enables the model to balance network stability and mobility efficiency.

Additionally, the exploration-exploitation ratio plays a critical role in training. Initially, the model explores different handover decisions to learn their impact, and as training progresses, it increasingly exploits the learned Q-values for optimal decisions. This ensures that the model generalizes well to real-world network variations, leading to adaptive, data-driven handover management that enhances overall network performance.

7. Future Work and Improvements

7.1 Possible Enhancements

Several modifications can be introduced to further improve the model's performance and accuracy. For Quality Signal Indicator (QSI) estimation, integrating latitude-longitude and user speed can make signal quality assessments more practical by considering geographical variations and mobility dynamics. In handover prediction, incorporating UE speed can make the model more adaptive to different mobility patterns, allowing for better prediction of handover events in both low- and high-speed scenarios. Additionally, fine-tuning the reward function parameters can help improve the balance between reducing unnecessary handovers and maintaining strong signal quality.

7.2 Scalability and Deployment

For real-world implementation, the model needs to be optimized for scalability, ensuring it can handle real-time decision-making across a large number of base stations. One approach is to integrate the model with Self-Organizing Networks (SONs), allowing autonomous network optimization. Edge computing solutions can also be leveraged to process handover decisions closer to the network edge, reducing latency. Further, efficient parallel processing and distributed learning techniques can be explored to scale the model across multiple network nodes while maintaining real-time performance.

7.3 Potential Research Directions

While our current work focuses primarily on reducing unnecessary handovers, future research can expand the optimization scope by incorporating additional parameters such as latency, throughput, and Quality of Service (QoS) metrics to enhance network efficiency. Additionally, deep reinforcement learning (DRL) can be explored to enable more sophisticated decision-making in complex environments. Another promising research direction is multi-agent Q-learning, where multiple base stations collaboratively optimize handover decisions, leading to improved network-wide efficiency. Further, context-aware handover strategies that factor in real-time user behavior, network congestion, and application requirements can be investigated to make handover management even more intelligent and adaptive.

8. Ethical Considerations and Sustainability

For wireless signal quality measurement, QSI, handover, and optimization, ethical considerations revolve around privacy, fairness, transparency, and regulatory compliance. Ensuring user data is protected is crucial, along with obtaining informed consent if real-world data is used. Measurements should not disrupt network performance for existing users, and methodologies must be transparent and reproducible to allow ethical scrutiny. Sustainability should be considered by optimizing network efficiency and reducing energy consumption. Security measures should be in place to prevent unauthorized access or misuse of sensitive data.

9. Conclusion

This study effectively explored the integration of multiple methodologies to optimize handover management in wireless communication networks. The Quality Signal Indicator (QSI) classification system, which categorized signal quality into four levels, provided a robust framework for assessing network performance across varying conditions. By utilizing percentile-based thresholding, the system ensured adaptive categorization, reflecting real-time network conditions.

The handover prediction component employed a set of ten supervised machine learning models, trained on real-world drive test data, to predict potential handover events. The models demonstrated strong performance, with particular emphasis on recall as the most critical metric in addressing the imbalance between handover and non-handover instances. This model was crucial in identifying handover events without missing critical instances, thereby ensuring service continuity.

For handover optimization, the implementation of Q-learning-based reinforcement learning effectively minimized unnecessary handovers, contributing to the stability of the network while enhancing mobility performance. By balancing the number of handovers and maintaining service quality, the Q-learning model demonstrated its ability to optimize network resources and improve overall efficiency.

In conclusion, the project successfully met its primary objectives of improving handover efficiency and network performance through the use of machine learning and reinforcement learning. The integration of these techniques resulted in a significant reduction in unnecessary handovers, thus improving signal quality and reducing network overhead. The outcomes of this study not only provide valuable insights into the optimization of handover processes but also lay the groundwork for further research into adaptive, self-optimizing networks. The approach presented here can contribute to the ongoing development of **5G** and beyond, where intelligent, dynamic handover management is essential for maintaining optimal user experience and network performance.

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59